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Research paper

Underwater side-scan sonar object detection with multi-dimensional feature enhancement and adaptive attention routing network

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ABSTRACT

Side-scan sonar (SSS) is widely utilized in seabed detection tasks, such as marine exploration and search and rescue operations. However, the balance between detection accuracy and efficiency is severely challenged by inherent SSS imagery artifacts. These artifacts include diffused edges, low signal-to-noise ratio (SNR), multi-scale variations, and complex backgrounds. Consequently, severe bottlenecks are imposed on underwater perception for resource-constrained platforms by the limitations of existing models. To address these issues, we propose a multi-scale edge-aware adaptive refinement network (MEAR-EffiNet). Specifically, a multi-scale edge refiner (MSERefiner) backbone is proposed to achieve multi-scale analysis and deep refinement of edge information from input features. The cross-stage macro-micro routing attention (C2MMRA) module is developed to capture long-range context. Moreover, a neck architecture is constructed by combining focused spatial adaptive local attention (FocSALA) and C3-based channel-frequency aggregation (C3CFA). Multi-dimensional attention and acoustic textures are leveraged to mitigate low SNR interference. Finally, a high-efficiency anchor-free regression-focused detection engine for noisy environments (HARDEN) is devised to model target edge localization uncertainty. Experiments on three challenging SSS datasets confirm that MEAR-EffiNet and outperforms state-of-the-art methods (e.g., YOLO/DETR series) in both accuracy (89.7% AP on SeabedObject-SSS). The proposed network has broad applications in complex SSS scenarios.

1. Introduction

Side-scan sonar (SSS) provides high-resolution imagery of the seafloor even in zero-visibility waters. Consequently, it is considered invaluable for diverse military and civilian applications. These applications include mine countermeasures (MCM), oceanographic mapping, and offshore oil and gas exploration (Steiniger et al., 2022). SSS is extensively employed for detecting submerged casualties, shipwrecks, and downed aircraft, particularly in underwater SAR missions (Reed et al., 2003). However, operator fatigue is caused by prolonged manual interpretation. The risk of missed targets is thereby increased. Consequently, the adoption of intelligent processing techniques is required for accurate, automated identification (Li et al., 2023a) (Wen et al., 2024).

SSS image segmentation methods were employed to identify anomalous seabed regions, such as highlighted objects or shadow areas. However, discriminative capability for specific target categories was typically lacking in their outputs (Aubard et al., 2025a). This limitation significantly reduced their utility for SAR missions. These missions necessitate the precise identification of distinct target types, such as submerged casualties, shipwrecks, or aircraft. Previous automated SSS target recognition research predominantly focused on the classification of narrowly defined mine-like objects (MLOs) (Yin et al., 2023). Reasonable performance was exhibited by these traditional approaches when dealing with consistent MLO shapes. However, they proved inadequate for complex non-MLO multi-category targets (Li et al., 2023b). Examples include human bodies, large shipwreck structures, or scattered aircraft debris. Complex targets characterized by diverse morphologies were not handled effectively. Consequently, simultaneous localization and differentiation of distinct target types remain a significant challenge (Grządziel, 2023).

Despite advancements, multi-category underwater SSS object detection remains significantly challenging (Preciado-Grijalva et al.,

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2022). Acoustic complexities, such as signal attenuation and multipath effects, cause severe edge blurring. Low-contrast targets are often indistinguishable from noisy backgrounds (Aubard et al., 2025b). Scale variations and occlusions further amplify these difficulties (Hoff et al., 2024). Deep learning, especially CNNs, was applied to SSS tasks (Agrawal et al., 2025). However, inherent limitations were presented when general-purpose CNNs were directly ported to multi-category SSS environments. Explicit mechanisms for severe edge blurring were often lacked by standard CNNs. Moreover, heterogeneous SSS multi-scale characteristics were not handled well by generic feature fusion (Aubard et al., 2024). Crucially, a stark accuracy-efficiency trade-off was faced by existing detectors. Accuracy was often compromised by lightweight models (e.g., YOLO) on low-SNR targets. Conversely, resource-constrained underwater deployment was hindered by the high computational cost of models like DETR. Therefore, a framework balancing high accuracy with real-time efficiency is urgently needed.

To address these challenges, we propose a multi-scale edge-aware adaptive refinement network (MEAR-EffiNet). A multi-scale edge refiner network (MSER-Net) is designed to overcome the limitations of generic detectors. Specialized mechanisms tailored for the unique characteristics of SSS data are integrated. The multi-scale edge refiner (MSERefiner) module is incorporated into the backbone. Blurred edges and multi-scale details are explicitly enhanced, thereby boosting perception for low-contrast targets. Moreover, a macro-micro routing mechanism is employed by the cross-stage macro-micro routing attention (C2MMRA) module. The focused spatial adaptive local attention (FocSALA) module is integrated into the multi-dimensional attention feature pyramid network (MDA-FPN) neck for noise suppression. Simultaneously, the C3-based channel-frequency aggregation with attention (C3CFA) module is utilized to leverage acoustic textures for feature fusion. Finally, an anchor-free design is utilized by the high-efficiency anchor-free regression-focused detection engine (HARDEN) detection head. Distribution Focal Loss (DFL) is integrated to precisely model localization uncertainty.

The main contributions of this work are described as follows.

(1) We propose an MSER-Net backbone. Ambiguous edges and multi-scale details are explicitly enhanced by its core MSERefiner module. Perception and detection capabilities for low-contrast targets and diffused contours are significantly boosted.
(2) We develop the C2MMRA module. Long-range contextual dependencies are efficiently captured by leveraging an innovative macro-micro routing mechanism. Micro-attention is guided to precisely focus on salient regions. This significantly enhances the discriminative power of deep semantic features, effectively addressing challenges posed by complex backgrounds and large, scattered targets.
(3) We design the MDA-FPN neck to incorporate the FocSALA module for noise suppression and spatial focusing. It is combined with the C3CFA module to leverage acoustic textures and spectral features. Multi-dimensional attention fusion is achieved by this dual-module approach. Multi-scale features are robustly integrated. Discrimination between acoustically similar targets and background clutter is enhanced.
(4) We devise HARDEN detection head. An anchor-free design is adopted for flexible handling of diverse target morphologies. Edge localization uncertainty is precisely modeled by integrating DFL. This enables highly efficient and accurate detection of various SSS targets, with its optimization guided by a comprehensive loss strategy.

2. Related work

2.1. Side scan Sonar image object detection

2.1.1. Side scan sonar imaging principle

SSS functions as an active acoustic detection system. Directional acoustic pulses are emitted from a transducer array toward the seabed. Backscattered echoes are recorded, as illustrated in Fig. 1. The time-of-flight of these echoes is directly proportional to the slant range, determining the target's distance. Topographic protrusions block acoustic transmission, creating acoustic shadows behind them where no signal is returned. Generally, high-intensity echoes are generated by hard, rough, or protruding surfaces. Conversely, low intensity is yielded by soft, flat, or shadowed regions. By continuously pulsing as the sonar platform advances, the system aggregates sequential scan lines to reconstruct a continuous two-dimensional seabed image. This technology is widely applied in hydrographic surveying, marine archaeology, and underwater search and rescue operations.

2.1.2. Underwater acoustic image object detection

Underwater exploration and target recognition are fundamentally supported by underwater acoustic imaging, particularly SSS. However, severe degradation is inherently suffered by SSS imagery. Speckle noise, low contrast, geometric distortion, and feature blurring are included. These issues are caused by acoustic propagation attenuation. Traditional signal processing and statistical models were primarily relied upon by early research. They were used for image enhancement or texture-based segmentation. However, the generalization capability required for complex seafloor environments was often lacking in these methods.

The mainstream approach was established by deep learning, specifically convolutional neural networks (CNNs) (Yang et al., 2025). Mature optical detection frameworks, such as YOLO and Faster R-CNN, were adapted to SSS data by researchers. However, bottlenecks were often encountered during the direct application of these models. Stringent demands were imposed on feature extraction and representation by the low signal-to-noise ratio (SNR). These demands were also contributed to by the distinct target boundaries in SSS images (Yang et al., 2024a) (Liu et al., 2024).

Consequently, a central research challenge is presented by the development of specialized network architectures. Acoustic degradation must be effectively mitigated while target structural information is precisely captured. Advanced exploration of edge perception and multi-scale contextual understanding mechanisms is necessitated.

2.2. Deep feature representation and fusion

2.2.1. Edge feature extraction and deep contextual understanding

In SSS imagery, inherent acoustic propagation factors and low contrast frequently cause severe target edge blurring, complicating precise extraction (Wu et al., 2025). Early gradient-based operators were noise-sensitive and failed on complex textures. However, superior robustness and adaptability were offered by CNN-based approaches. Architectural enhancements were leveraged by recent advancements. These included multi-scale fusion and attention mechanisms. Alternatively, traditional edge detection was integrated with deep features to significantly improve target discriminability.

Deep semantic information and long-range spatial contexts are crucial for interpreting complex scenes. Scattered targets are also recognized through these features. However, these long-range dependencies were difficult to capture due to the limited receptive fields of traditional CNNs. Arbitrary spatial relationships could be modeled by global attention mechanisms. Yet, prohibitive computational costs were often incurred on high-resolution feature maps. Consequently, context was captured with reduced overhead by efficient variants. Examples included sparse, window, and pyramidal attention. However, features were often processed by these methods only at a single, final stage

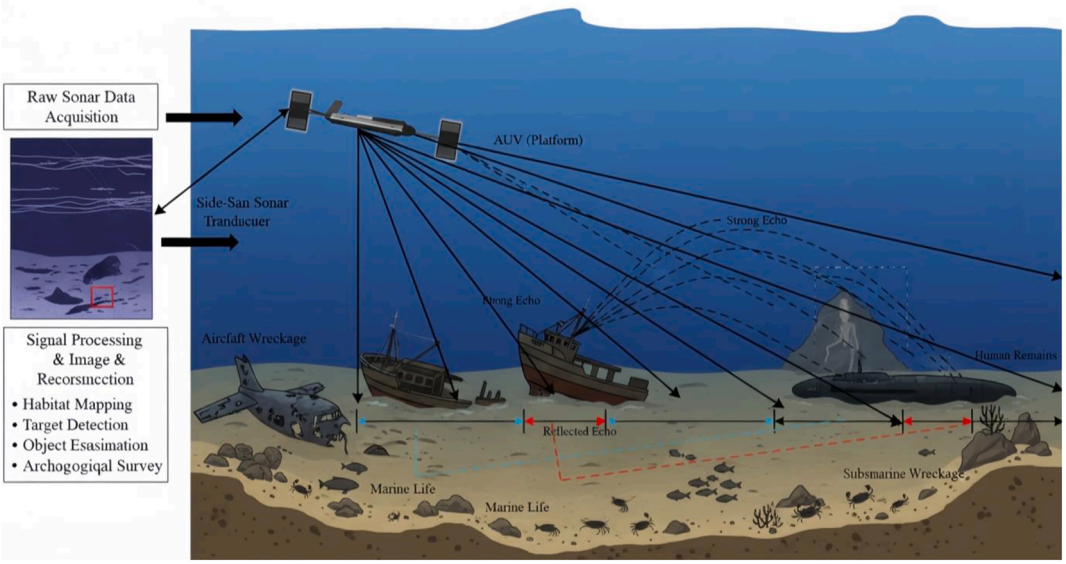

Fig. 1. The principle of side-scan sonar.

(Wang et al., 2025a). The necessity of cross-scale information fusion was overlooked. This fusion is vital for detecting variable-sized targets in complex SSS backgrounds.

2.2.2. *Feature pyramid networks and multi-scale fusion*

The feature pyramid network (FPN) served as a foundational architecture for multi-scale detection. Deep semantics were fused with shallow spatial details. However, noise propagation failed to be suppressed by its simple element-wise summation. Feature complementarity was not fully leveraged, particularly in low signal-to-noise ratio (SNR) SSS imagery. To mitigate this, bottom-up pathways were introduced by advanced architectures like the path aggregation network (PANet) (Liu et al., 2018). Feature flow was strengthened. Meanwhile, learnable weights were employed by the weighted Bi-directional FPN (BiFPN) (Tang, 2022) for optimized feature balancing. Moreover, critical regions were adaptively focused on by integrating attention mechanisms or dynamic reorganization strategies. Signal-to-noise discrimination in complex acoustic backgrounds was thereby enhanced.

Ultimately, detection performance is dependent on the efficient translation of these refined features into final predictions. Simultaneously, a critical bottleneck remains for practical deployment on resource-constrained platforms. These platforms include autonomous underwater vehicles (AUVs) (Wu et al., 2025). This bottleneck is constituted by the computational overhead of sophisticated fusion networks.

2.3. *Precise model prediction and lightweight design*

At the prediction stage, the design paradigm was shifted from rigid anchor-based methods. Tedious hyperparameter tuning was demanded by those methods. Instead, superior adaptability to morphologically diverse targets was offered by flexible anchor-free architectures (e.g., FCOS, YOLOX) (Ponzini et al., 2025) (Shen et al., 2025). Bounding box regression was evolved from direct coordinate prediction to advanced probability distribution modeling. This was done to address the blurred boundaries inherent in SSS imagery. Localization accuracy was enhanced by these methods, inspired by distribution focal loss (DFL). Discrete probability distributions of boundary distances were predicted. They were guided to concentrate near the ground truth. Localization uncertainty was effectively mitigated by this approach. This proved critical for SSS images where precise boundaries were often obscured by acoustic blurring.

Moreover, lightweight model design is prioritized given the stringent resource constraints of underwater platforms (Chen et al., 2024a). Optimization of backbones, feature fusion networks, and prediction heads is entailed. Efficient techniques were used, such as depthwise separable convolutions, group convolutions, and structural reparameterization (Wang et al., 2024a). Model parameters and computational load (FLOPs) must be significantly reduced without compromising performance. Consequently, efficient prediction mechanisms (e.g., anchor-free designs) are synergistically integrated with lightweight architecture. Robust, real-time SSS object detection is ensured by this critical combination.

3. Method

3.1. *Overall network architecture*

Inherent challenges in underwater SSS imagery are addressed by the proposed MEAR-EffiNet. These challenges include diffused target contours, blurred edges, multi-scale variations, complex background interference, and low signal-to-noise ratio (SNR). The structure of MEAR-EffiNet is illustrated in Fig. 2. Four key components are included in MEAR-EffiNet: MSER-Net, MDA-FPN, C2MMRA, and HARDEN. Edge-aware adaptive refinement is facilitated by this architecture across multiple scales. Consequently, robust noise suppression and spatial focusing are achieved, making it suitable for efficient edge deployment.

The processing pipeline is initiated with the extraction of deep feature representations by the backbone. Specifically, blurred edges caused by acoustic artifacts are enhanced by MSER-Net. Simultaneously, multi-scale morphological details are captured at critical layers. The C2MMRA module is positioned in the deeper layers. An attention mechanism is utilized by this module to capture long-range contextual dependencies and refine high-level semantics. Complex scene understanding is thereby facilitated. A hierarchical feature set rich in both structural detail and semantic depth is yielded synergistically by these components.

These features are subsequently fed into the MDA-FPN neck. A lightweight focused spatial adaptive attention mechanism is integrated with a channel-frequency aggregation unit within the neck. Information flow is optimized by this design through the discriminative enhancement and alignment of multi-scale features. Finally, the refined features are processed by the HARDEN detection head. Adopting an anchor-free design coupled with DFL, HARDEN flexibly adapts to diverse target

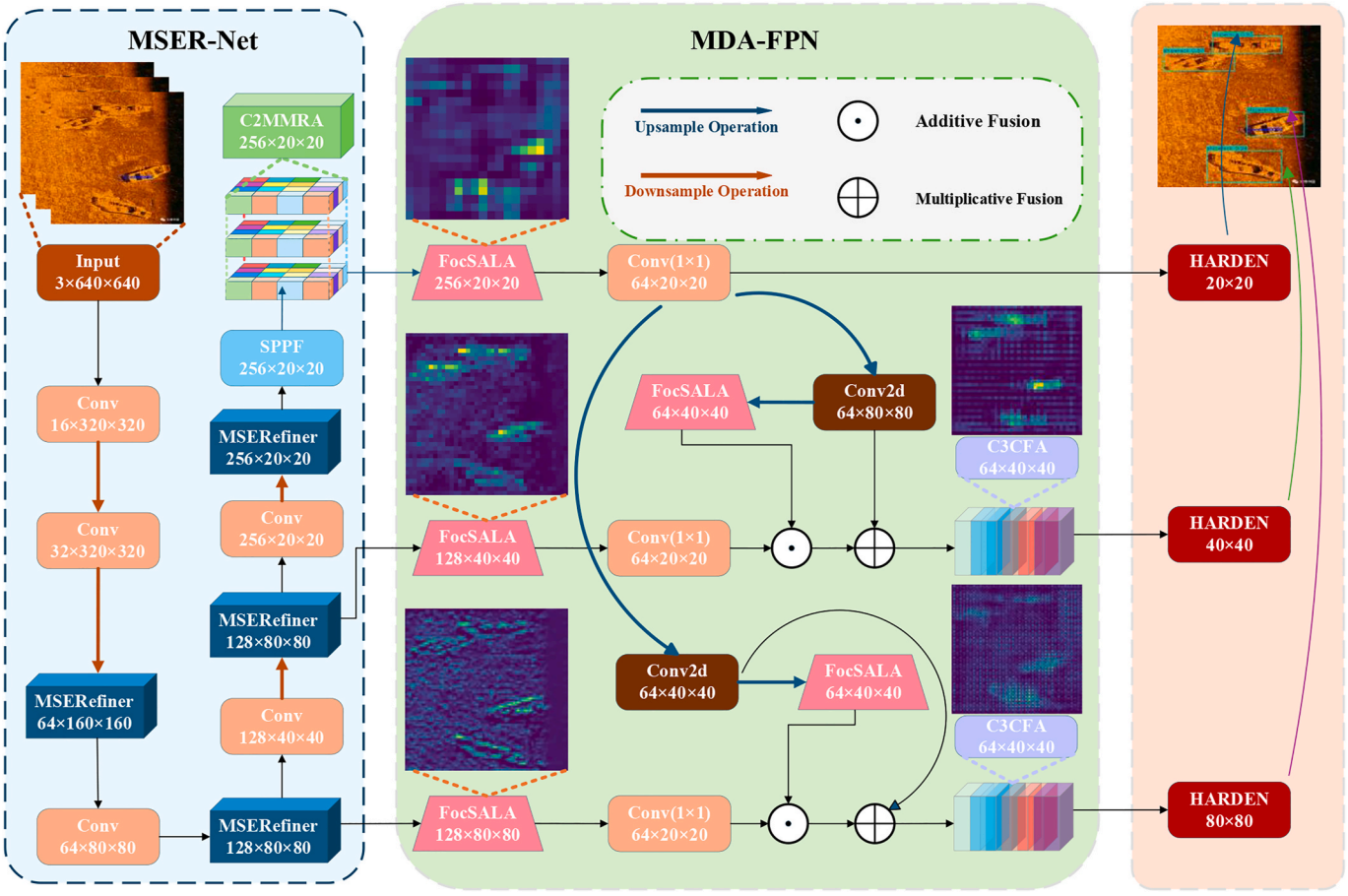

Fig. 2. The overall structure of MEAR-EffiNet. It comprises four key components: the MSERefiner backbone for robust feature extraction, the C2MMRA module for deep semantic feature enhancement, the MDA-FPN neck for multi-scale feature fusion and refinement, and the HARDEN detection head for precise object localization and classification. Convolution (Conv) with filter size 3 × 3; SPPF, spatial pyramid pooling fast layer.

morphologies and effectively mitigates localization uncertainty arising from indistinct edges.

3.2. MSER-Net: multi-scale edge refinement feature fusion backbone network

Severe challenges are faced by object detection in underwater SSS imagery. This is especially true for targets with complex scales and varied morphologies (e.g., drowned victims, shipwreck debris, small aircraft). These challenges stem from significant variations in contour scale and inherent edge blurriness (Cheng et al., 2022). Critically, acoustic reflection characteristics highly similar to the complex seafloor background are exhibited by many underwater targets. Examples include sediment-covered shipwrecks or drowned victims in environments like sand ripples or rugged terrain. Extremely low target-background contrast and inherent edge blurriness are caused by this similarity. Consequently, precise identification is made exceedingly difficult for traditional feature extraction methods.

To address these issues, we propose a multi-scale edge refiner (MSERefiner) module as a core component of the backbone. This module is designed specifically to mitigate SSS degradation. Effective multi-scale feature extraction is employed to capture diverse target sizes. Simultaneously, advanced enhancement techniques are utilized to sharpen blurred contours. Unlike general-purpose visual models that merely stack standard operations, a hierarchical, progressive feature enhancement strategy is adopted by MSERefiner. As illustrated in the backbone configuration, modules are deployed at layers 2, 4, 6, and 8. The perception of multi-scale information and critical edge details is continuously strengthened across different depths and receptive fields. Feature discriminability is effectively enhanced by this tailored approach. The network is enabled to distinguish targets from the background even when edges are indistinct. Overall detection performance is thereby significantly improved.

3.2.1. MSERefiner module structure and principle

To address the challenges of blurred edges and low contrast inherent in SSS imagery, we propose the MSERefiner module (Fig. 3). This module executes a dual-phase strategy of multi-scale dispersion followed by fusion to achieve deep refinement of edge information. Specifically, preliminary local contextual features are first extracted from an input feature map $F_{in} \in \mathbb{R}^{W\times H\times C}$ via a local 3 × 3 convolution. A foundation for subsequent processing is established. To accommodate varying target sizes and potential sediment occlusion, the module employs multiple parallel branches. In each branch i, feature representations at different spatial resolutions are captured from the original input. An adaptive average pooling layer with distinct bin parameters is employed:

$$F_{pool_i} = \text{AdaptiveAvgPool2d}_{bin_i}(F_{in}) \tag{1}$$

where F_{in} is the input feature map. $\text{AdaptiveAvgPool2d}_{bin_i}(\cdot)$ denotes the adaptive average pooling operation with a specific bin parameter bin_i. F_{pool_i} is the resulting pooled feature map in branch i.

Subsequently, a 1 × 1 convolution is applied to each pooled feature to facilitate cross-channel information interaction. This is followed by a 3 × 3 group convolution (Conv_g) to extract spatial features:

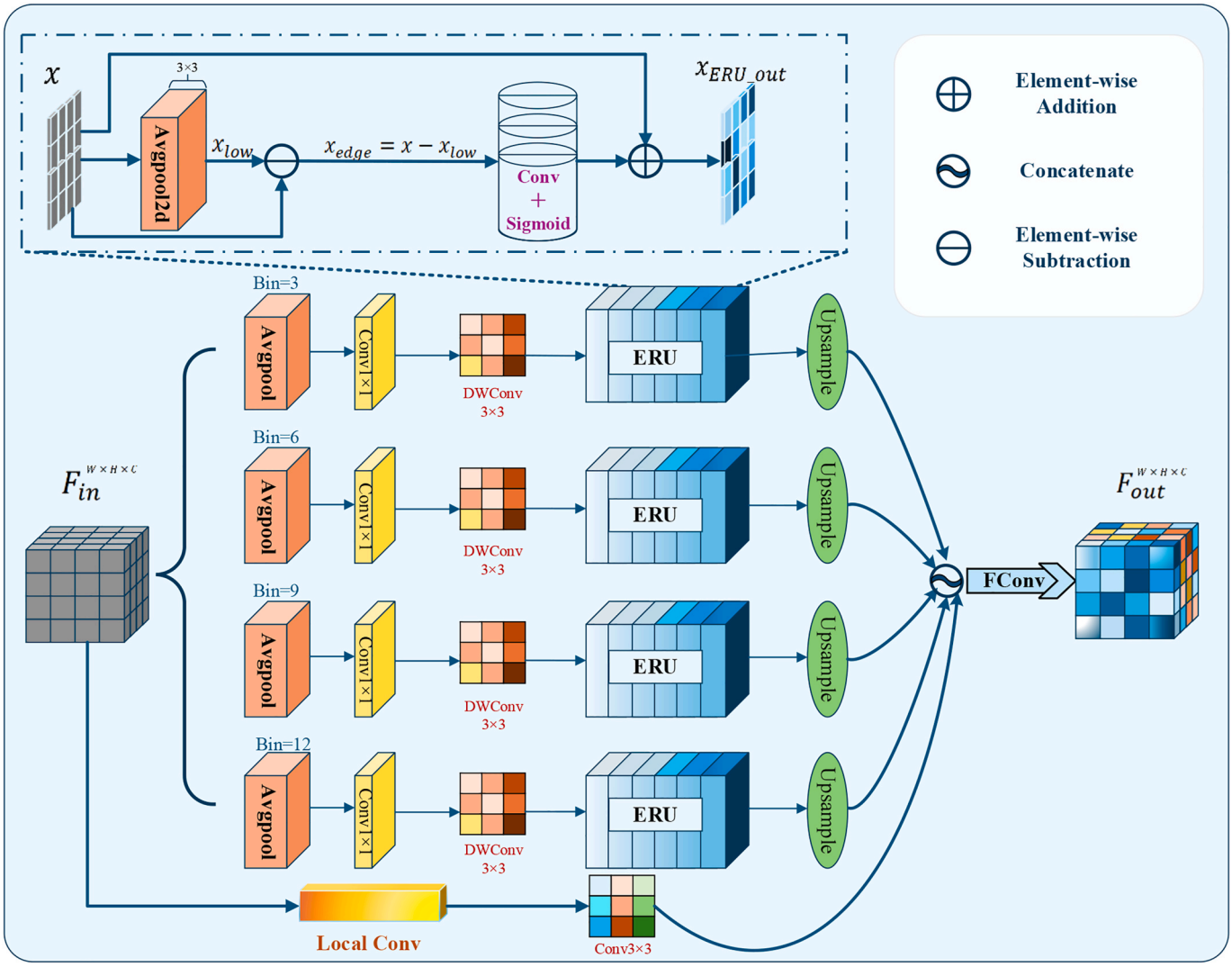

Fig. 3. The structure of the MSERefiner module. A key internal component is the edge refinement unit (ERU), which contributes to robust feature representation and noise reduction.

$$F_{scale_{feat_i}} = \mathrm{Conv}_{g,i}\left(\mathrm{Conv}_{1\times1,i}\left(F_{pool_i}\right)\right) \tag{2}$$

where $F_{scale_feat_i}$ is the extracted scale-specific feature in branch i, $\mathrm{Conv}_{1\times1,i}(\cdot)$ represents the 1×1 convolution, and $\mathrm{Conv}_{g,i}(\cdot)$ signifies the group convolution operation1. A critical innovation within this architecture is the integration of the EdgeRefine unit (ERU) into each branch. SSS speckle noise is manifested as high-frequency signals, similar to target edges. However, differentiation is effectively achieved by the MSERefiner through the processing of features after preliminary encoding. Operations are performed on feature maps where raw pixel-level noise has been filtered by preceding layers. Consequently, the enhancement of structural boundaries with semantic consistency is selectively focused on by the ERU. Random noise fluctuations are not amplified. Finally, the refined features from all branches are upsampled via bilinear interpolation. They are concatenated and fused through a final convolutional layer to produce the enhanced output F_{out}.

3.2.2. EdgeRefineUnit design

The EdgeRefineUnit (ERU) is explicitly designed as the core component of the MSERefiner module. High-frequency boundary signals are isolated from low-frequency background fluctuations. Consequently, the issue of indistinct contours caused by signal attenuation is addressed. The process is initiated by applying a smoothing operation to the input feature map x. Low-frequency information is extracted by an average pooling layer. This process is expressed as:

$$x_{low} = \mathrm{AvgPool2d}(x) \tag{3}$$

where x is the input feature map, and x_{low} denotes the extracted low-frequency component. The global trend of the feature distribution is captured by this component. Examples include the uniform seabed background.

Subsequently, this low-frequency component is subtracted from the original input. High-frequency signals corresponding to rapid intensity changes at boundaries are isolated:

$$x_{edge} = x - x_{low} \tag{4}$$

where x_{edge} represents the preliminary high-frequency edge features. The saliency of potentially weak boundary signals is enhanced against the background by this differential operation.

To further distinguish valid target edges from high-frequency speckle noise, the extracted x_{edge} undergoes a non-linear refinement process. It is fed into a convolutional layer followed by a Sigmoid activation function. Adaptive weights are learned to amplify structurally significant edges while incoherent noise responses are suppressed:

$$x_{refined_{edge}} = \mathrm{Sigmoid}\left(\mathrm{Conv}\left(x_{edge}\right)\right) \tag{5}$$

where $\mathrm{Conv}(\cdot)$ is the convolution operation, $\mathrm{Sigmoid}(\cdot)$ is the activation function, and $x_{refined_edge}$ is the refined edge feature map. Normalized response values in the range $(0,1)$ indicate the confidence of edge existence.

Finally, these enhanced cues are integrated back into the original feature stream by a residual connection. Semantic context is thereby preserved:

$$x_{ERU_{out}} = x + x_{refined_{edge}} \tag{6}$$

where $x_{ERU_{out}}$ is the final output feature map of the ERU, x is the original input feature map. This additive mechanism allows the network to autonomously balance the contribution of edge details. The enhancement complements the original features rather than introducing artifacts.

3.3. MDA-FPN structure and principle

Traditional FPNs often prove inadequate for processing SSS imagery. Such imagery is characterized by intense noise, low contrast, and significant target scale variations. Stringent demands are imposed on neck networks by the inherent complexities of acoustic imaging. Capabilities transcending conventional scale fusion are necessitated. Precise extraction and enhancement of faint target signals are thereby ensured.

To address these challenges, we propose the MDA-FPN. As illustrated in Fig. 2, a distinct architectural evolution over traditional FPNs is presented by MDA-FPN. Two specialized attention and aggregation units are strategically embedded: FocSALA and C3CFA. Fine-grained adaptive modulation and deep information aggregation are facilitated by this design across spatial, channel, and frequency dimensions. Feature flow and interaction within the pyramid are effectively redefined.

This approach diverges from the direct upsampling and element-wise summation of standard FPNs. An adaptive spatial focusing mechanism is integrated via the FocSALA module prior to fusion. Responses in potential target regions are enhanced by this mechanism. Diffuse seafloor reverberation and noise are simultaneously suppressed. Feature extraction in low SNR environments is significantly improved.

Moreover, interactive channel-transposed attention (iCTA) and dual frequency aggregation FFN (DFFN) are employed by the C3CFA module. Deep feature refinement is achieved. Unique channel maps (C-Maps) and Spatial Maps (S-Maps) are leveraged by iCTA for cross-dimensional gating to enhance expressiveness. Simultaneously, frequency-domain cues are exploited by DFFN. Targets are distinguished from backgrounds with similar acoustic intensities but distinct textures. Precise alignment and high-quality integration of high-level semantics and low-level details are ensured by this multi-stage, multi-dimensional processing.

3.3.1. FocSALA module structure and principle

The FocSALA module is illustrated in Fig. 4. It is designed as a pivotal unit within the neck network. The local expressive capability of feature maps is dynamically augmented via a parameter-efficient spatial attention mechanism. Its architectural philosophy centers on capturing contextual dependencies along both horizontal and vertical axes, thereby achieving adaptive "focusing" on salient regions. Specifically, the core focused spatial attention (FSA) unit initiates processing by performing spatial dimension compression on the input feature map $x \in \mathbb{R}^{B\times C\times H\times W}$ to encode contextual information.

Two parallel adaptive average pooling layers are employed by the FSA unit to achieve this. Global average pooling is performed along the height (H) and width (W) dimensions, respectively yielding:

$$x_h = \mathrm{AdaptiveAvgPool2d}(x, (\mathrm{None}, 1)) \in \mathbb{R}^{B\times C\times H\times 1} \tag{7}$$

$$x_w = \mathrm{AdaptiveAvgPool2d}(x, (1, \mathrm{None})) \in \mathbb{R}^{B\times C\times 1\times W} \tag{8}$$

where x_h is obtained by pooling the input x across its width dimension (to an output size of $H\times 1$), thereby encoding height-wise context. Similarly, x_w is obtained by pooling across the height dimension (to an output size of $1\times W$), encoding width-wise context. This compression strategy effectively condenses 2D spatial information into 1D vectors, significantly mitigating computational overhead and establishing the foundation for the module's lightweight design.

To model local dependencies and generate attention weights, x_h and x_w are reshaped into tensors of $\mathbb{R}^{B\times C\times H}$ and $\mathbb{R}^{B\times C\times W}$ to accommodate 1D convolutional processing. These two reshaped tensors share a 1D convolutional module $f_{1D}(\cdot)$. This module consists of a 1D convolutional layer (nn. Conv1d with kernel size $K_c = 7$ and padding $P_c = 3$ to maintain length), a group normalization layer (nn.GroupNorm, with G groups), and a Sigmoid activation function $\sigma(\cdot)$. This configuration captures local spatial patterns while stabilizing training, mapping the output to the $(0,1)$ range to formulate the attention weights:

$$att_h = \sigma(\mathrm{GroupNorm}(\mathrm{Conv1d}(x_h.\mathrm{reshape}(B,C,H)))) \in \mathbb{R}^{B\times C\times H} \tag{9}$$

$$att_w = \sigma(\mathrm{GroupNorm}(\mathrm{Conv1d}(x_w.\mathrm{reshape}(B,C,W)))) \in \mathbb{R}^{B\times C\times W} \tag{10}$$

where att_h and att_w are the generated attention weights for the height and width dimensions, respectively. $\mathrm{Conv1d}(\cdot)$ represents the shared 1D convolutional module, $\mathrm{GroupNorm}(\cdot)$ is the group normalization layer, and $\sigma(\cdot)$ denotes the Sigmoid activation function.

Finally, the learned spatial attention weights are applied to the original input feature map x according to the configuration parameter flag. To this end, att_h and att_w are reshaped back to dimensions $\mathbb{R}^{B\times C\times H\times 1}$ and $\mathbb{R}^{B\times C\times 1\times W}$ respectively. When the module is configured for feature enhancement mode (flag = True), adaptive focusing and enhancement of the original features are achieved. Element-wise multiplication (Hadamard product $\odot$) is performed on the original feature map x with

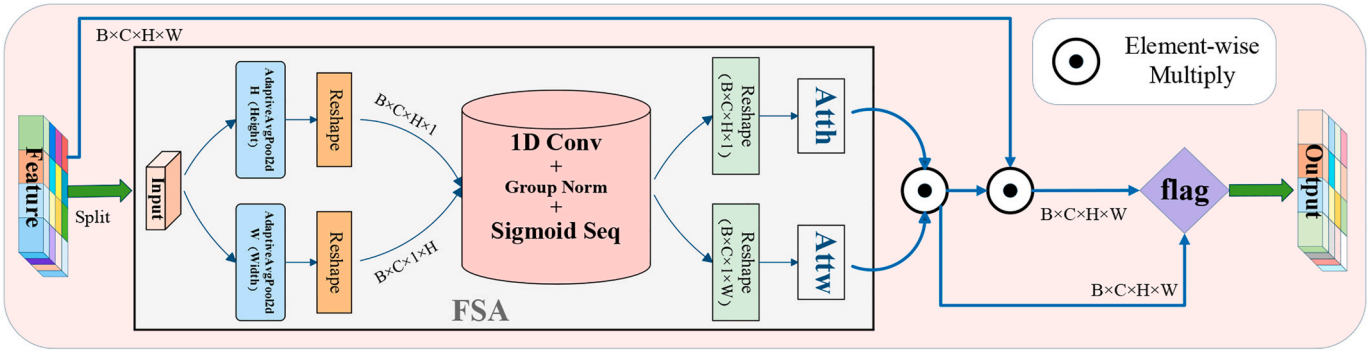

Fig. 4. The structure of the FocSALA module. As a key component of the neck network, FocSALA is designed for noise-robust spatial focusing and multi-scale feature fusion. Its core functionality is driven by the embedded FSA Attention.

its learned weights along the height and width dimensions:

$$x_{out} = x \odot att_h.\text{reshape}(B, C, H, 1) \odot att_w.\text{reshape}(B, C, 1, W) \tag{11}$$

where x_{out} is an enhanced feature map.

This operation suppresses background noise while emphasizing salient spatial locations, achieving a "focused spatial adaptive" effect. Conversely, when the module is configured for fusion guidance mode (flag = False), the module outputs the direct product of the directional attention weights:

$$x_{out} = att_h.\text{reshape}(B, C, H, 1) \odot att_w.\text{reshape}(B, C, 1, W) \tag{12}$$

where x_{out} serves as a pure spatial focusing map.

In this part, $x_{out} \in \mathbb{R}^{B\times C\times H\times W}$ is employed as a pure spatial saliency map. High-priority regions are indicated in both horizontal and vertical directions. The prioritization of information flow during feature fusion is guided.

3.3.2. C3CFA: deep feature refinement and discriminative enhancement for SSS imager

Illustrated in Fig. 5, the C3CFA module facilitates deep feature interaction, information aggregation, and discriminative refinement. It serves as a core engine for detecting complex underwater targets. Channel-frequency aggregation (CFA) units are stacked by the module using the efficient C3 structure. Feature stream splitting and merging are utilized to promote gradient propagation and feature diversity. The interactive channel-transposed attention (iCTA) module is designed, drawing inspiration from the FreqFormer (Dai et al., 2024). It addresses bottlenecks in discerning subtle target features and handling background complexity. A residual design is adopted by each CFA unit. The iCTA module and the dual frequency aggregation feed-forward network (DFFN) are serially concatenated. Comprehensive optimization is achieved across channel, spatial, and frequency-aware dimensions. For an input feature $x_{cfa_in} \in \mathbb{R}^{B\times C'\times H\times W}$ to the CFA unit, the processing flow is formulated as sequential transformation and residual learning:

$$x'_{cfa} = \text{iCTA}\left(\text{LN}\left(x_{cfa_{in}}\right)\right) \tag{13}$$

$$x_{cfa_{out}} = x'_{cfa} + \text{DropPath}\left(\text{DFFN}\left(\text{LN}\left(x'_{cfa}\right)\right)\right) \tag{14}$$

where $x_{cfa_{in}}$ is the input feature to the CFA unit, $\text{LN}(\cdot)$ denotes the layer normalization operation, and $\text{iCTA}(\cdot)$ is the interactive channel-transposed attention module. The term x'_{cfa} is the intermediate feature after the iCTA block. $\text{DFFN}(\cdot)$ is the dual frequency aggregation FFN, $\text{DropPath}(\cdot)$ is a stochastic depth regularization method, and $x_{cfa_{out}}$ is the final output of the CFA unit.

The iCTA module constitutes the perceptual core, transcending traditional self-attention limitations. For the dimension-transformed input feature $x_{ln} \in \mathbb{R}^{B\times N\times C'}$ (where $N = HW$), iCTA first generates Query (Q), Key (K), and Value (V) matrices through three independent linear projections $W_Q, W_K, W_V \in \mathbb{R}^{C'\times C'}$: $Q = x_{ln}W_Q$, $K = x_{ln}W_K$, $V = x_{ln}W_V$. Subsequently, scaled dot-product attention is computed under a multi-head mechanism to capture long-range dependencies:

$$F_{attn} = \text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}\cdot\tau}\right)V \tag{15}$$

where $Q, K, V \in \mathbb{R}^{B\times N\times C'}$ represent the projected feature subspaces, $d_k = C'/N_{heads}$ is the scaling factor preventing gradient vanishing, and τ is a learnable temperature parameter regulating attention sharpness. Simultaneously, to compensate for the lack of local inductive bias, a parallel depthwise separable convolution path processes the Value features $V' \in \mathbb{R}^{B\times C'\times H\times W}$ (reshaped from V) to generate local convolutional features $F_{conv} = \text{DWConv}(V') \in \mathbb{R}^{B\times C'\times H\times W}$. Crucially, iCTA introduces a cross-dimensional interaction mechanism where a Channel Map ($M_C \in \mathbb{R}^{B\times C'\times 1\times 1}$) and a Spatial Map ($M_S \in \mathbb{R}^{B\times 1\times H\times W}$) are generated to produce gating signals via Sigmoid activation. These signals multiplicatively modulate the features of the opposing path, enabling the module to adaptively suppress noise while reinforcing target-consistent signals.

Following the iCTA is the dual frequency aggregation feed-forward network (DFFN), serving as another core component of the CFA unit. It aims to enhance feature discriminability from a more fundamental signal characteristic level. While traditional FFNs primarily perform non-linear mapping between channels, DFFN processes feature with frequency awareness through its core FrequencyGate (FG) unit $f_{FG}(\cdot, H, W)$. It is important to note that acoustic texture is strictly governed by the operating frequency. Acoustic texture and speckle noise patterns vary significantly across samples (coarse to fine-grained) since the dataset aggregates multi-source data (400–1600 kHz). The FrequencyGate unit f_{FG} is specifically designed to handle this heterogeneity by capturing and utilizing these frequency domain cues. Formally, for an input $x_{dffn_in} \in \mathbb{R}^{B\times N\times C'}$, DFFN processes it via a linear transformation W_1, an activation function $\text{act}(\cdot)$, and the FrequencyGate unit f_{FG}:

$$x_{hidden} = \text{act}\left(x_{dffn_{in}}W_1\right) \in \mathbb{R}^{B\times N\times C_{hidden}} \tag{16}$$

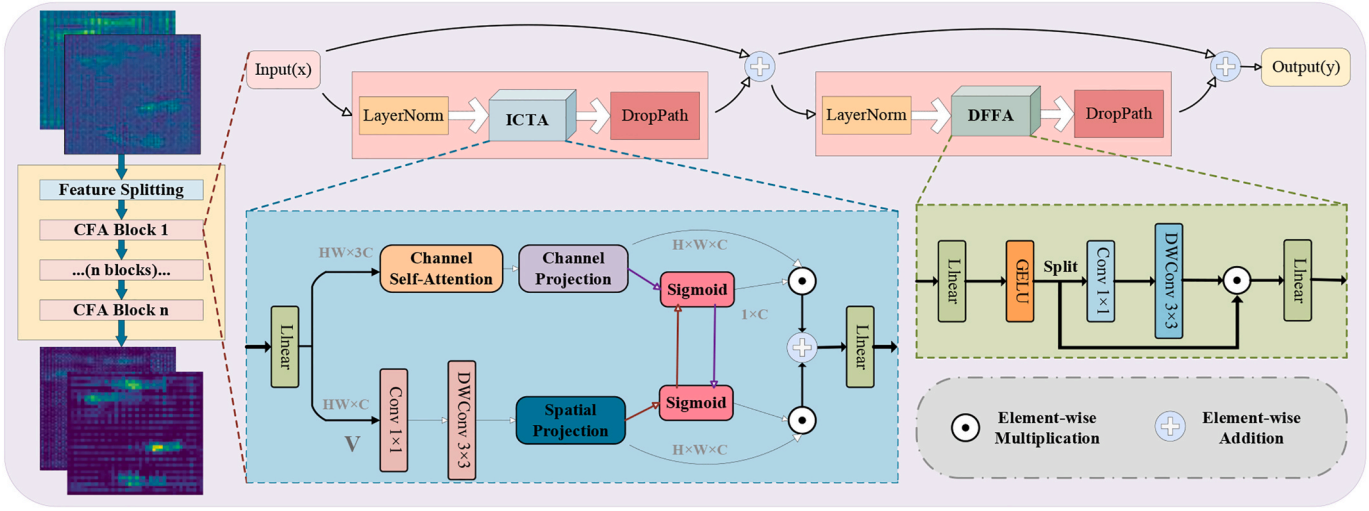

Fig. 5. The structure of the C3CFA module. Located in the neck network after the FocSALA module, this module enhances feature discriminability through inter-channel transformation attention (ICTA) and dynamic frequency feature aggregation (DFFA).

$$x_{gated} = f_{FG}(x_{hidden}, H, W) \in \mathbb{R}^{B\times N\times C'_{hidden}} \tag{17}$$

$$\mathrm{DFFN}(x_{dffn_{in}}) = x_{gated}W_2 + \mathrm{Dropout}(x_{gated}W_2) \tag{18}$$

where $W_1 \in \mathbb{R}^{C'\times\gamma C'}$ and $W_2 \in \mathbb{R}^{\gamma C'\times C'}$ are learnable weight matrices with an expansion ratio γ, and act$(\cdot)$ denotes the activation function. The intermediate feature $x_{hidden} \in \mathbb{R}^{B\times N\times\gamma C'}$ is modulated by the FrequencyGate f_{FG}. $f_{FG}(\cdot, H, W)$ denotes the FrequencyGate unit, which processes features in the frequency domain given the spatial dimensions (H, W) to produce the gated feature x_{gated}. Dropout$(\cdot)$ is used for regularization. Critically, to maintain computational efficiency and phase consistency, f_{FG} is implemented in the spatial domain rather than using explicit spectral transformations. Specifically, we utilize a Depthwise Separable Convolution with a 3×3 kernel and groups $=$ C to function as a learnable spatial filter bank. Rapid intensity changes—corresponding to high-frequency acoustic textures—are spatially encoded by this design. They are distinguished from low-frequency background signals. Representations effectively differentiating targets are learned by the module through the dynamic weighting of these frequency-aware spatial cues.

3.4. C2MMRA: macro-micro routing enhancement for deep semantic features

Object detection in underwater SSS imagery is hindered by morphologically diverse targets and complex acoustic backgrounds. Global semantic modeling capabilities of deep features are challenged. The P5 stage feature map contains condensed semantic abstractions. However, it often lacks the long-range context necessary for accurate recognition due to the inherent locality of traditional convolutions. While standard global attention ($O((HW)^2)$) is computationally intractable and linear attention ($O(HW)$) often fails to capture the spatial heterogeneity of SSS data. The C2MMRA module is designed to address these limitations.

As illustrated in Fig. 6, the C2MMRA is strategically deployed at the 10th layer of the backbone, immediately following the SPPF module. Drawing inspiration from the dynamic sparse attention mechanism in BiFormer (Zhu et al., 2023). We design the feature extraction paradigm by reformulating the bi-level routing strategy into a computationally efficient C2-based bottleneck architecture. Unlike the original implementation in Vision Transformers, a local context enhancement (LCE) path is explicitly incorporated. Fine-grained acoustic textures are preserved while long-range dependencies are modeled. The C2MMRA module outputs the enhanced high-level feature F_{MMRA_out} through an embedded macro-micro routing attention (MMRA) mechanism.

The MMRA unit processes the input feature map $X \in \mathbb{R}^{N\times C\times H\times W}$. First, a 1×1 convolution performs linear projection to obtain Query (Q), Key (K), and Value (V) tensors:

$$[Q, K, V] = \mathrm{Split}_{\mathrm{dim}=1}(\mathrm{Conv}_{1\times 1}(X)) \tag{19}$$

where $Q, K, V \in \mathbb{R}^{N\times C\times H\times W}$.The first stage of MMRA is macro-region routing. The feature map is divided into $n_{win}\times n_{win}$ non-overlapping regions, with region dimensions $(h_r, w_r) = (H/n_{win}, W/n_{win})$. For Q and K, average pooling is applied within each region, yielding macro representatives:

$$Q_{macro} = \mathrm{AvgPool}_{h_r,w_r}(Q.\mathrm{detach}(\cdot)) \in \mathbb{R}^{N\times C\times n_{win}\times n_{win}} \tag{20}$$

$$K_{macro} = \mathrm{AvgPool}_{h_r,w_r}(K.\mathrm{detach}(\cdot)) \in \mathbb{R}^{N\times C\times n_{win}\times n_{win}} \tag{21}$$

where Q_{macro} and K_{macro} are the macro-level representations of Q and K, respectively. The detach$(\cdot)$ function is used to block gradient flow at this stage.

The macro-region affinity matrix A_{macro} is computed, reflecting semantic relationships between regions:

$$A_{macro} = Flatten(Q_{macro}) \times Flatten(K_{macro})^T \in \mathbb{R}^{N\times(n_{win}^2)\times(n_{win}^2)} \tag{22}$$

where A_{macro} represents the inter-region affinity scores. They are calculated by the matrix product of flattened Q_{macro} and K_{macro}.

Based on A_{macro}, a dynamic pruning strategy selects the top-k most relevant regions, constructing a sparse connectivity graph Idx_{macro}:

$$Idx_{macro} = \mathrm{TopK}_k(A_{macro}, \mathrm{dim} = -1) \in \mathbb{Z}^{N\times(n_{win^2})\times k} \tag{23}$$

where the $\mathrm{TopK}_k(\cdot)$ function selects the indices of the k regions with the highest affinity along the last dimension, generating the sparse connectivity graph Idx_{macro}.

Convolutional locality is effectively overcome by this macro-routing mechanism. Long-range correlations are established at a controllable cost. A global perspective is provided. This is crucial for discerning large-scale underwater targets.

The second stage, Micro-Token Attention, executes fine-grained interactions guided by Idx_{macro}. To optimize efficiency, the attention calculation for each query token is strictly confined to key-value tokens within its routed regions:

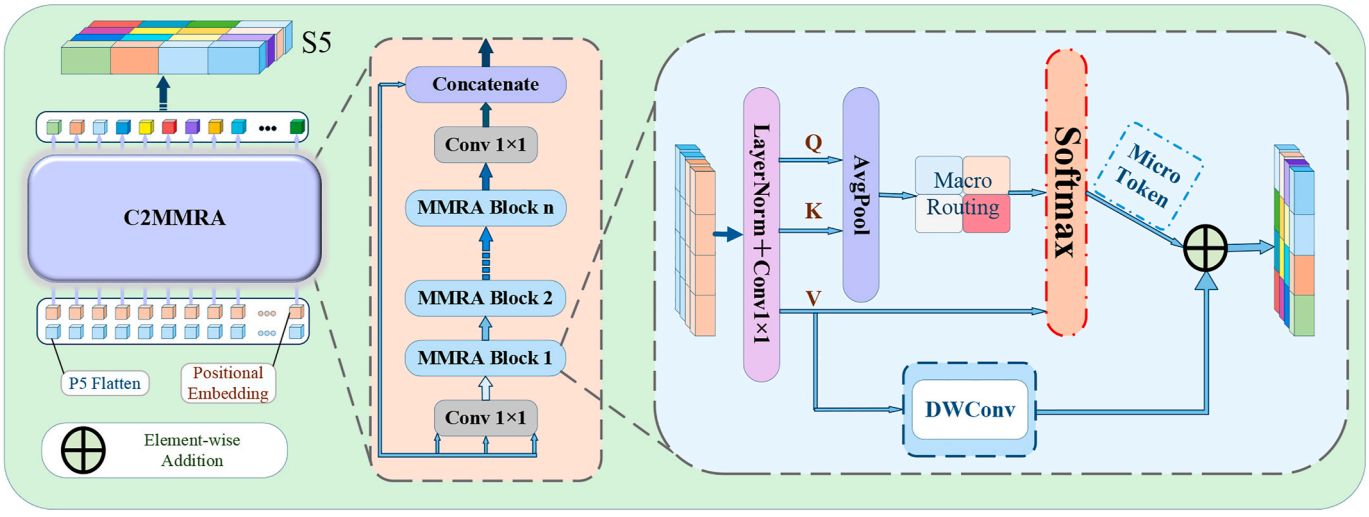

Fig. 6. The structure of the C2MMRA module. This module, which locates in the backbone network after MSERefiner, enhances deep semantic feature discriminability by capturing long-range context and focusing on key regions. Its core is the macro-micro routing attention (MMRA).

$$\text{Attention}(Q_{sub}, K_{routed}, V_{routed}) = \text{Softmax}\left(\frac{Q_{sub}K_{routed}^{T}}{\text{scale}}\right)V_{routed} \tag{24}$$

where K_{routed} and V_{routed} are key-value tokens aggregated based on Idx_{macro}. Simultaneously, to address the potential loss of local details in attention mechanisms, we design a parallel LCE path:

$$V_{LCE} = \text{Conv}_{DW}(\text{kernel} = side_d wconv)(V) \tag{25}$$

where V_{LCE} is the locally context-enhanced feature obtained by applying a depthwise separable convolution to the original value tensor V.

The MMRA unit's final output fuses the micro-attention output with V_{LCE}, followed by a 1×1 convolutional projection:

$$Output_{\text{MMRA unit}} = \text{LinearProjection}(\text{Attention}(Q, K_{\text{routed}}, V_{\text{routed}}) + V_{LCE}) \tag{26}$$

where LinearProjection$(\cdot)$ denotes the final 1×1 convolutional layer used to integrate the information from the attention mechanism and the local context enhancement.

P5 features are iteratively refined by stacking MMRA units within the C2 bottleneck. Positioned after SPPF, the module leverages multi-scale receptive fields. Noise is suppressed and core semantics are strengthened. A functionally complementary pipeline is formed with MSERefiner and SPPF. The backbone is endowed with powerful, discriminative representation capabilities for SSS imagery.

3.5. HARDEN: an anchor-free regression-focused noise detection head and its loss optimization strategy

3.5.1. Module structure and principle

The detection head serves as the critical terminal for predicting object classes and bounding boxes. This follows the feature refinement and fusion performed by the backbone (MSER-Net) and neck (MDA-FPN). Its efficacy hinges on the synergy between architectural design and loss optimization. Drawing inspiration from the decoupled head architecture of YOLOv8 (Jocher et al., 2023), we devise the HARDEN detection head, as illustrated in Fig. 7. While leveraging the effective decoupled paradigm of YOLOv8, we develop a structural evolution tailored for resource-constrained underwater devices. A lightweight group convolution stem is designed to replace standard heavy convolutions. This modification significantly reduces computational complexity and parameter volume, enabling faster inference speeds essential for real-time underwater detection.

The HARDEN module processes multi-scale feature maps $\{P_l\}_{l=0}^{L-1} \in \mathbb{R}^{N\times C_l\times H_l\times W_l}$ from the neck. For a given input P_l, the customized stem module, Stem_l, employs two sequential 3×3 group convolutions. Specifically, to balance efficiency and feature representation, the group number g is dynamically set to $x//16$, where x denotes the input channels. This mechanism partitions channels into 16 distinct groups for independent processing, substantially lowering computational costs compared to standard convolutions ($g = 1$) used in prior works. The refined feature map P_l' is formulated as:

$$P_l' = \text{Stem}_l(P_l) \tag{27}$$

where P_l' is fed into two parallel 1×1 convolutional branches: $\text{Conv}_{1\times1}^{\text{box},l}$ predicts the discrete probability distribution for the bounding box (with $4 \times \text{reg_max}$ channels), and $\text{Conv}_{1\times1}^{\text{cls},l}$ predicts the class logits (with nc channels). Their outputs are concatenated to form the raw prediction O_l for scale l:

$$O_l = \left[\text{Conv}_{1\times1}^{\text{box},l}(P_l'), \text{Conv}_{1\times1}^{\text{cls},l}(P_l')\right] \in \mathbb{R}^{N\times(4\cdot\text{reg_max}+\text{nc})\times H_l\times W_l} \tag{28}$$

where $\text{Conv}_{1\times1}^{\text{box},l}(\cdot)$ and $\text{Conv}_{1\times1}^{\text{cls},l}(\cdot)$ are the 1×1 convolutions for bounding box regression and classification, respectively. *reg_max* defines the range for the discretized boundary distances, and *nc* is the number of object classes.

During inference, multi-scale O_l are integrated and parsed by an Anchor-Free decoder combined with DFL, generating the final detection results.

3.5.2. Loss function and training process

HARDEN's performance is intrinsically linked to its loss function design. The loss strategy is coupled with the anchor-free output, optimizing classification, regression, and label assignment to tackle SSS challenges.

For classification, SSS imagery presents severe class imbalance and many hard samples. HARDEN outputs class logits. The base classification loss employs binary cross-entropy with logits loss (BCEWithLogitsLoss) $\mathscr{L}_{cls_base}$:

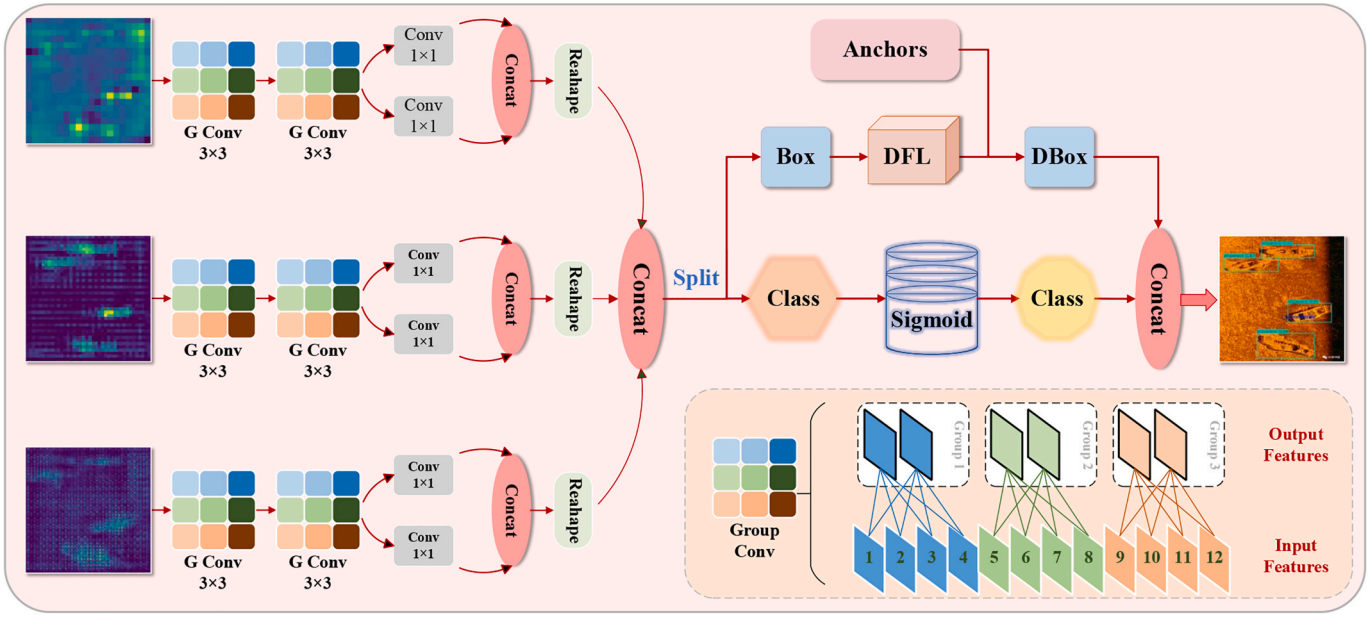

Fig. 7. The structure of the HARDEN module. For a given input, the customized stem module, Stem_l, employs two sequential 3×3 group convolutions.

$$\mathscr{L}_{cls_{base}}(x,y) = -[y\log(\sigma(x)) + (1-y)\log(1-\sigma(x))] \tag{29}$$

where x represents the raw logits output by the network, y is the ground-truth label, and $\sigma(\cdot)$ is the Sigmoid function.

To address hard samples, the classification loss $\mathscr{L}_{cls}$ integrates focal loss, which focuses on difficult samples via the $(1-p_t)^{\gamma}$ modulating factor:

$$L_{cls}(p_t) = -\alpha_t(1-p_t)^{\gamma}\log(p_t) \tag{30}$$

where $p_t = \sigma(x)$ is the predicted probability, $\gamma \geq 0$ is the focusing parameter, and $\alpha_t \in [0,1]$ is the balancing factor. This loss enables HARDEN to effectively learn to distinguish low-contrast, blurred targets.

For bounding box regression, to cope with blurred edges, HARDEN predicts a discrete probability of distribution for boundary distances. It employs DFL for optimization, encouraging the probability distribution to concentrate around the true value y_{gt} at y_k, y_{k+1}. The predicted continuous distance $\widehat{y}$ is the expected value of this probability distribution:

$$\widehat{y} = \sum_{i=0}^{reg_max-1} P(y_i)\cdot y_i \tag{31}$$

where $P(y_i)$ is the model's predicted probability that the boundary distance is y_i.

The DFL loss $\mathscr{L}_{dfl}$ penalizes distribution deviation:

$$\mathscr{L}_{dfl}(P(y_k),P(y_{k+1})) = -\Big(\big(y_{k+1}-y_{gt}\big)\log(P(y_k)) + \big(y_{gt}-y_k\big)\log(P(y_{k+1}))\Big) \tag{32}$$

where y_{gt} is the ground-truth boundary distance, which lies between the discrete points y_k and y_{k+1}. $P(y_k)$ and $P(y_{k+1})$ are the model's predicted probabilities for these two points.

DFL is complemented by an IoU-based loss (e.g., GIoU). The geometric overlap between the predicted box B_{pred} and ground truth B_{gt}:

$$\mathscr{L}_{iou} = 1 - \mathrm{GIoU}\big(B_{pred},B_{gt}\big) = 1 - \left(\mathrm{IoU}\big(B_{pred},B_{gt}\big) - \frac{\big|C\big(B_{pred}\cup B_{gt}\big)\big|}{|C|}\right) \tag{33}$$

where C is the smallest convex hull enclosing both boxes. $\mathscr{L}_{iou}$ and $\mathscr{L}_{dfl}$ synergistically overcome the challenge of indistinct contours.

During training, we utilize the TaskAlignedAssigner (TAL) to dynamically select top-k positive samples based on a metric $t = s^{\alpha}\cdot u^{\beta}$, where s is classification confidence and u is IoU. This ensures high-quality alignment between classification and localization tasks.

The total training objective $\mathscr{L}_{total}$ constitutes a weighted sum:

$$\mathscr{L}_{total} = w_{cls}\mathscr{L}_{cls} + w_{box}\mathscr{L}_{iou} + w_{dfl}\mathscr{L}_{dfl} \tag{34}$$

where w_{cls}, w_{box}, and w_{dfl} are the weights used to balance the classification, IoU, and DFL losses.

The weights $w_{cls}, w_{box}, w_{dfl}$ (corresponding to self.hyp.cls, self.hyp.box, self-hyp.dfl) balance the optimization process and are tuned to addressing the specific difficulties of SSS data.

4. Experiment

4.1. Dataset and implementation details

4.1.1. Dataset

In this section, we evaluate the performance of the proposed network using the SeabedObject-SSS dataset (gx-123, 2024). This collection encompasses complex imaging scenarios and diverse noise characteristics, providing a rigorous testing environment. The dataset comprises 986 high-resolution Side-scan sonar (SSS) images, detailed in three categories: aircraft, human, and shipwreck.

Regarding the "Human" category, clarification on target authenticity is necessary. The samples in this subset primarily consist of anthropomorphic mannequins deployed in controlled underwater search and rescue (SAR) exercises. We acknowledge a potential domain gap between these rigid surrogates and real-world subjects, which may exhibit varying degrees of distinct deformation due to environmental factors. However, the use of surrogates is justified by the principles of underwater acoustics. Specifically, these surrogates effectively simulate the fundamental acoustic physical properties of the human body, particularly the acoustic impedance mismatch with the water medium and the resulting backscattering characteristics. By training on these samples, the model learns to capture robust acoustic representations and reflectivity features that are physically consistent with real targets, ensuring practical transferability to actual SAR operations.

These images feature varying water depths and, critically, aggregate data acquired from side-scan sonars operating at multiple frequencies, including 400 kHz, 600 kHz, 900 kHz, and 1600 kHz. This multi-frequency composition introduces significant variability in acoustic texture and resolution, ranging from coarse-grained speckle noise at lower frequencies to fine-grained textures at ultra-high frequencies, as illustrated in the bottom row (e-h) of Fig. 8.

However, a critical challenge with such open-source sonar datasets is the frequent absence of raw acoustic logs, which contain precise metadata like sensor altitude and slant range. To address this limitation and ensure the physical validity of the experimental assessment, an image-derived acoustic verification method (Padial et al., 2014) is employed to qualitatively categorize the acquisition conditions. We analyze the acoustic shadows and backscatter intensity to infer the geometric and environmental diversity. As shown in the top row(a-d) of Fig. 8, the dataset exhibits targets with significantly varying acoustic shadow lengths (highlighted by red circles in Fig. 8(a) and (b)); based on the physics of sonar imaging, elongated shadows observed in the samples indicate low grazing angles, while minimal shadows correspond to high grazing angles. Moreover, the variations in mean pixel intensities (indicated by blue circles in Fig. 8(c) and (d)) simulate different time-varied gain settings and seabed sediment reverberation levels. By validating these acoustic properties through the above method, we confirm that the dataset provides a sufficiently diverse physical distribution for training robust detection models, effectively simulating real-world underwater application scenarios.

To further validate the generalization capability of the proposed network, we also utilize two additional datasets: SeabedObjects-KLSG (Huo et al., 2020) and Optical-Ship (Wei et al., 2024). The SeabedObjects-KLSG dataset was constructed by Huo et al. For this study, only the first version of this dataset is employed, which contains 518 images covering the categories of Aircraft, Shipwreck, Fish, and other targets, along with a subset of target-free seabed background images. The images in this dataset have a resolution of 640 × 640 pixels. We utilize these target-free images as negative samples. Regarding the Optical-Ship dataset, it was compiled by Wei et al. via style transfer from optical aerial images. Specifically, the sonar-semi version is used in this paper, which includes 1114 images containing the Shipwreck category. We utilize this dataset specifically as a generic cross-domain generalization test. It is important to note that as synthetic data generated via transfer learning, these style-transferred images inherently lack authentic acoustic physical properties, serving primarily to evaluate algorithmic robustness under domain shifts.

To ensure a fair evaluation of the proposed network across different target types and datasets, all three datasets (SeabedObject-SSS, SeabedObjects-KLSG, and Optical-Ship) are randomly split into training, validation, and testing sets at an 8:1:1 ratio, respectively.

4.1.2. Implementation protocols and hyperparameter settings

The comprehensive network configuration and implementation guidelines are detailed in Table 1, encompassing annotation protocols, data augmentation strategies, early stopping criteria, input dimensions,

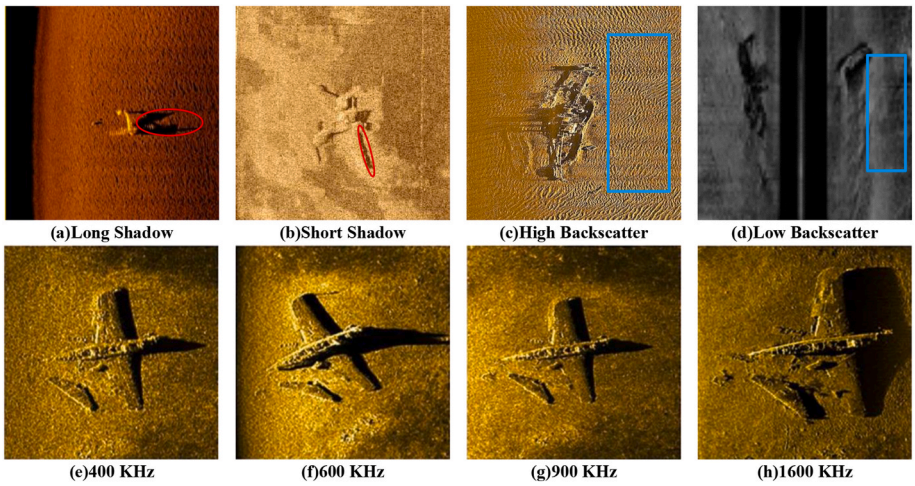

Fig. 8. Acoustic diversity features of the sonar dataset. (a) Target with an elongated acoustic shadow, indicating a low grazing angle. (b) Target with a compact shadow, indicating a high grazing angle. (c) Sample exhibiting high backscatter intensity. (d) Sample exhibiting low backscatter intensity. Bottom row (e-h): variability in acoustic texture due to operating frequencies.

Table 1
Comprehensive implementation protocols and hyperparameters.

Category	Parameter	Specification/Value
1. Training setup	Hardware	Intel i7-13700KF/NVIDIA RTX 3090 (24 GB)
	Software	Windows 11 Pro/Python 3.8.19/ PyTorch 2.1.1/CUDA 12.5
2. Deployment setup	Hardware	Rockchip RK3588s2 SoC (NPU: 6 TOPS)
	Framework	Ubuntu 20.04/RKNN-Toolkit2/ONNX
	Quantization	INT8
3. Data protocol	Input size	Fixed: 640×640
	Augmentation strategy	Mosaic: Disabled ($p = 1.0$) Geometric: Random Flip ($p = 0.5$), Scale (0.5-1.5)
4. Model architecture	Global scaling (Scale-n)	Width factor: 0.25 (Base channels), Depth factor: 0.50
	Internal expansion (e)	$e = 0.25$ (Shallow Layers), $e = 0.5$ (Deep Layers)
5. Training dynamics	Batch	16
	Epochs	500
	Patience	50
	Optimizer	AdamW (YOLO Default)
6. MSERefiner	Multi-scale bins	List: $[3, 6, 9, 12]$ (Adaptive pooling scales)
	Low-pass filter	Spatial AvgPool ($k = 3; s = 1, p = 1$)
7. C3CFA	iCTA attention	Heads: 8; QKV bias: False
	Channel projection	Dilated Conv ($k = 7, d = 3, p = 9$)
	DFFN expansion	Ratio $r = 4.0$ (Hidden dim $= 4 \times C_{in}$)
	FreqGate	Depthwise Conv ($k = 3, s = 1, p = 1, g = C_{in}$)[7]

and channel configurations. Additionally, it explicitly specifies the core implementation parameters for the frequency-domain processing components utilized in the MSERefiner and C3CFA modules.

To further substantiate the practical deployability of the proposed architecture on resource-constrained underwater platforms, real-world inference speed tests are conducted using the Rockchip RK3588s2 embedded computing board, as visualized in Fig. 9. This system-on-chip (SoC) features a high-performance octa-core 64-bit CPU architecture (comprising four Cortex-A76 and four Cortex-A55 cores) and an embedded ARM Mali-G610 GPU. Crucially for edge AI tasks, a proprietary neural processing unit (NPU) is integrated, delivering up to 6 TOPS of computing power with support for mixed-precision operations (INT4/INT8/FP16). The device is equipped with 8 GB RAM and 64 GB storage. For the deployment pipeline, the trained PyTorch models (.pt) are first exported to the ONNX standard and subsequently converted to the RKNN format using the RKNN-Toolkit2 within an Ubuntu 20.04 environment. INT8 post-training quantization is employed to optimize memory bandwidth and latency. The final inference speed, measured in frames per second (FPS), is evaluated directly on the RK3588s2 target hardware to assess the model's real-time capability in actual underwater edge scenarios.

Under this rigorous deployment protocol, MEAR-EffiNet achieves an inference speed of 69 FPS. This quantitative result effectively substantiates the model's lightweight efficiency. Furthermore, it confirms the architecture's capability to satisfy real-time requirements on resource-constrained edge hardware. The specific data is detailed in Table 2.

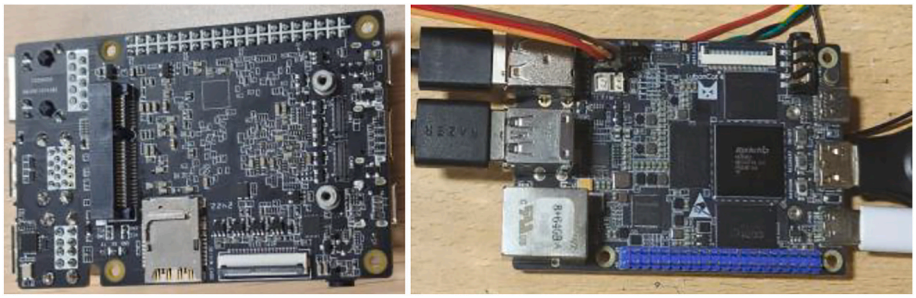

Fig. 9. Rockchip RK3588s2 embedded edge computing platform.

Table 2
Comparison with state-of-the-art models on SeabedObject-SSS. Bold and underline fonts are the highest and the second highest, respectively.

Model	*Param.*(*M*)	*FLOPs*(*G*)	*AP*	AP_{50}	AP_{75}	AP_S	AP_M	AP_L	*FPS*	*Precision*	*Recall*
Real − time Object Detectors											
YOLOv5-N (bib_'releasesReleases)	2.5	7.1	80.4	47.6	48.0	29.4	49.5	47.6	72	84.2	71.4
YOLOv6-N (Li et al., 2022)	4.2	11.8	83.0	50.8	54.0	45.0	55.1	55.8	**76**	86.8	73.2
YOLOv8-N (Jocher et al., 2023)	3.0	8.1	83.4	50.6	54.1	30.1	52.4	53.0	65	85.3	79.3
YOLOv9-T (Wang et al., 2025b)	**1.9**	7.6	84.9	53.7	54.6	43.4	55.9	58.6	70	85.5	82.4
YOLOv10-N (Wang et al., 2024b)	2.7	8.2	78.9	50.5	53.6	36.9	54.8	55.2	67	76.6	73.2
YOLOv11-S (GitHub)	9.4	21.3	84.4	53.7	**55.6**	52.5	56.6	57.0	36	85.4	76.8
YOLOv12-N (Tian et al., 2025)	2.5	6.3	82.5	49.9	48.5	45.5	52.0	51.8	68	84.2	76.5
Hyper-YOLO-N (Feng et al., 2025)	3.6	9.5	87.3	52.9	55.4	55.0	53.3	58.1	54	94.3	79.2
End − to − end Object Detectors											
DAB-DETR (Liu, 2022)	43.7	65.3	79.5	51.6	53.1	26.5	45.3	55.4	-	82.7	75.3
DINO (Zhang et al., 2022)	47.6	179.0	83.4	48.2	50.8	25.4	46.8	56.1	-	84.3	74.2
RT-DETR-R18 (Zhao et al., 2024)	28.3	42.7	82.1	50.1	52.3	23.8	44.2	56.2	-	79.2	87.5
Salience-DETR (Hou et al., 2024)	48.9	305.0	80.7	45.4	45.9	27.7	42.8	54.5	-	76.6	**94.2**
Relation-DETR (Hou et al., 2025)	19.2	56.7	79.2	45.5	47.5	22.6	44.5	54.6	-	77.2	93.3
D-FINE-S (Peng et al., 2024)	10.1	24.8	84.6	49.8	51.4	31.5	49.8	57.2	-	85.1	85.2
Baseline											
YOLOv11-N (GitHub)	2.6	6.3	85.0	53.7	52.4	25.0	**58.2**	54.6	64	91.2	77.5
Ours	2.3	**4.7**	**89.7**	**56.1**	55.3	**57.6**	57.1	**59.2**	69	**94.5**	81.0

4.2. Comparative experiments

4.2.1. Comparative experiments on network architecture

To comprehensively validate the effectiveness of the proposed MEAR-EffiNet, we conduct a systematic benchmarking on the SeabedObject-SSS dataset against mainstream real-time detectors (YOLO series) and end-to-end detectors (DETR series). The quantitative results are summarized in Table 2. Adopting YOLOv11-N as the baseline due to its leading balance of speed and accuracy (85.0% AP) among existing methods, MEAR-EffiNet achieves a superior AP of 89.7% (a relative improvement of $\Delta AP = 4.7\%$), while its parameter counts and FLOPs decrease by $\Delta P = 0.3M$ and $\Delta F = 1.6G$. Notably, its accuracy surpasses even the larger YOLOv11-S model by 5.3%, demonstrating exceptional hardware efficiency suitable for complex underwater environments. Fig. 10 provides a qualitative comparison, where MEAR-EffiNet distinguishes highly confusable targets (e.g., aircraft wreckage vs. shipwrecks) with more precise localization and higher confidence, effectively suppressing background interference.

Comparison with real-time detectors: We evaluate MEAR-EffiNet against a spectrum of lightweight models, including YOLOv5-N through Hyper-YOLO-N. MEAR-EffiNet exhibits distinct advantages in detection accuracy, outperforming the state-of-the-art YOLOv12-N and

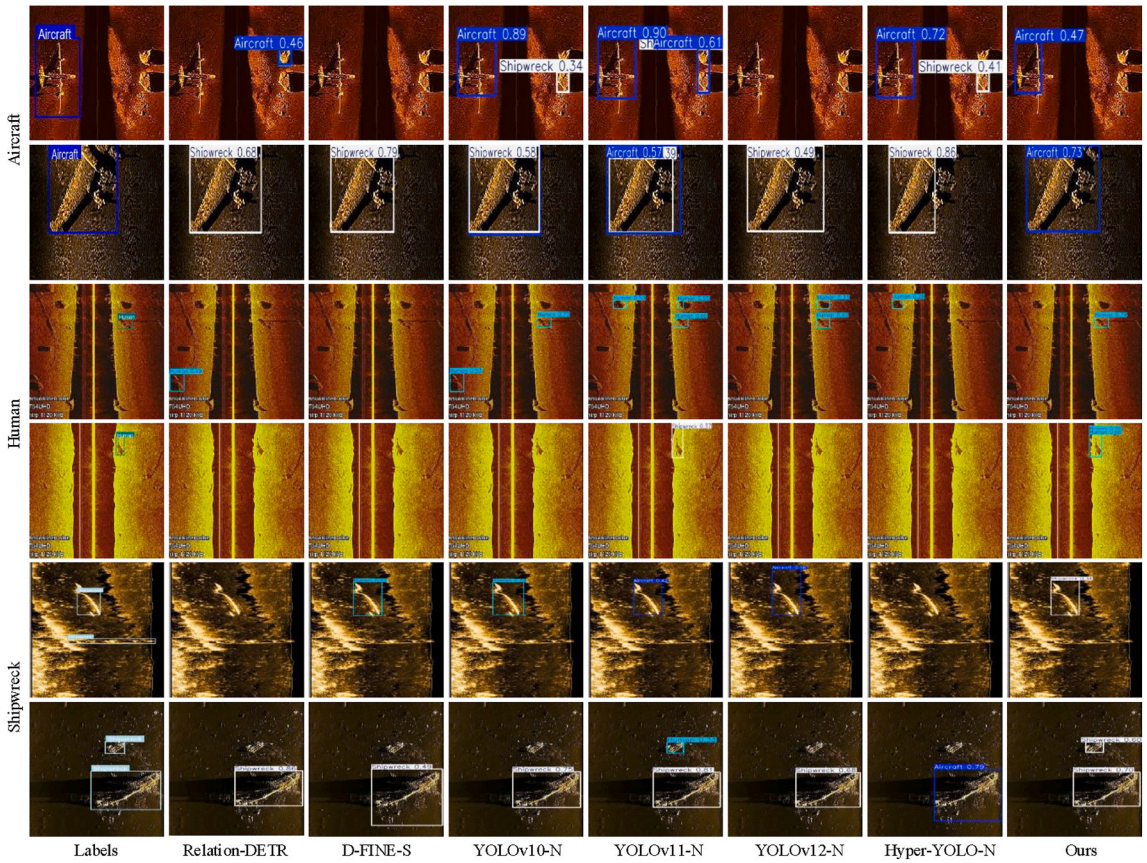

Fig. 10. Visualization results of representative object detection methods for three object categories on the SeabedObject-SSS dataset.

Hyper-YOLO-N by 8.7% and 2.7% AP, respectively. Crucially, this performance gain is achieved with a lighter architectural footprint; for instance, compared to YOLOv12-N, parameter counts and FLOPs are reduced by 8.0% and 25.4%, respectively, highlighting significant advantages over Hyper-YOLO-N.

Comparison with end-to-end detectors: Against DETR-series models, MEAR-EffiNet consistently maintains a significant accuracy edge while operating with substantially lower computational costs, satisfying the stringent constraints of edge deployment. Even compared to D-FINE, the top-performing end-to-end model, MEAR-EffiNet improves AP by 6.0% while drastically minimizing resource overhead. These results confirm that MEAR-EffiNet achieves an optimal trade-off between performance and efficiency.

To visually illustrate this dual advantage, Fig. 11(a) and (b) plot the AP-Params and AP-FLOPs curves, respectively; MEAR-EffiNet consistently occupies the upper-left quadrant, signifying higher accuracy at lower resource consumption.

MEAR-EffiNet achieves an inference speed of 69 frames per second (FPS) on the RK3588s2, which comfortably exceeds the real-time processing requirement ($>$ 30 FPS) for underwater video streams. It is confirmed by this result that excellent efficiency is maintained by the architecture, despite the inclusion of sophisticated frequency-aware modules, making it fully applicable to actual underwater edge platforms. End-to-end transformer-based models (e.g., DETR) are deliberately excluded from this specific edge speed comparison, as they are rendered inherently unsuitable for deployment on low-power edge devices by their excessive parameter counts and computational demands, making a direct speed comparison irrelevant in this context. Moreover, Table 2 shows comprehensive performance metrics, including Precision, Recall, and specific AP scores for small (AP_S), medium (AP_M), and large (AP_L) objects.

4.2.2. Comparative experiments on MSER-Net

To systematically evaluate the effectiveness of the proposed MSER-Net, we replace the backbone network with six popular lightweight/1 architectures: ConvNeXtV2 (Woo et al., 2023), EfficientVit (Liu et al., 2023b), FasterNet (Chen et al., 2023), LSKNet (Li et al., 2023c), PKINet (Cai et al., 2024), and RepViT (Wang et al., 2024c). Under identical conditions regarding dataset, training strategy, and hyperparameters, model parameters (Param.), computational load (FLOPs), and detection performance (AP/AP50/AP75) are uniformly recorded.

The quantitative results in Table 3 highlight MSER-Net's superior efficiency-accuracy trade-off. It achieves an AP of 89.7% with merely 2.3 M parameters and 4.7 G FLOPs. Notably, it outperforms the second-best candidate, PKINet (5.8M), by an additional 1.4% in AP, despite the latter possessing a significantly larger parameter footprint. This underscores MSER-Net's exceptional feature representation capability within a compact budget. Moreover, Table 3 shows comprehensive performance metrics, including Precision, Recall, and specific AP scores for small (AP_S), medium (AP_M), and large (AP_L) objects.

To qualitatively assess feature responsiveness, Fig. 12 presents HiResCAM (Draelos and Carin, 2021) visualizations comparing MSER-Net against other backbones on targets such as “Aircraft” and “Shipwreck.” While competing models (e.g., YOLOv11N, ConvNeXtV2) often exhibit scattered activations or blurred boundaries, MSER-Net generates highly concentrated and intense responses along target edges and key semantic regions. This phenomenon confirms that the integration of the multi-scale edge refinement strategy significantly sharpens contour delineation and semantic capture, directly contributing to its superior detection robustness in complex underwater scenarios.

4.2.3. Quantitative evaluation of noise suppression and edge enhancement in the MSERefiner module

To quantitatively evaluate the effectiveness of the MSERefiner module in differentiating target boundaries from background speckle noise, we introduce two key metrics: signal-to-noise ratio (SNR) and contrast-to-coise ratio (CNR). These metrics are defined as:

$$SNR = 20\log_{10}\left(\frac{\mu_t}{\sigma_b}\right) \tag{35}$$

$$CNR = \frac{|\mu_t - \mu_b|}{\sigma_b} \tag{36}$$

where u_t and u_b represent the mean feature intensity of the target and background regions, respectively, and σ_b denotes the standard deviation of the background noise. We conduct this evaluation on the representative mid-level feature maps (Layer 6), which balance spatial resolution with semantic abstraction. As presented in Table 4, the experimental results indicate a substantial improvement in feature quality after passing through the MSERefiner module. Specifically, the SNR increase from 7.84 dB to 8.92 dB (+1.07 dB), demonstrating that the module effectively suppresses the noise floor without amplifying high-frequency speckle interference. More significantly, the CNR achieve a dramatic improvement of 52.5% (rising from 0.97 to 1.48). This sharp increase confirms that the MSERefiner successfully widens the statistical gap between target edges and the surrounding seabed texture, ensuring that valid boundary signals are prioritized over background clutter.

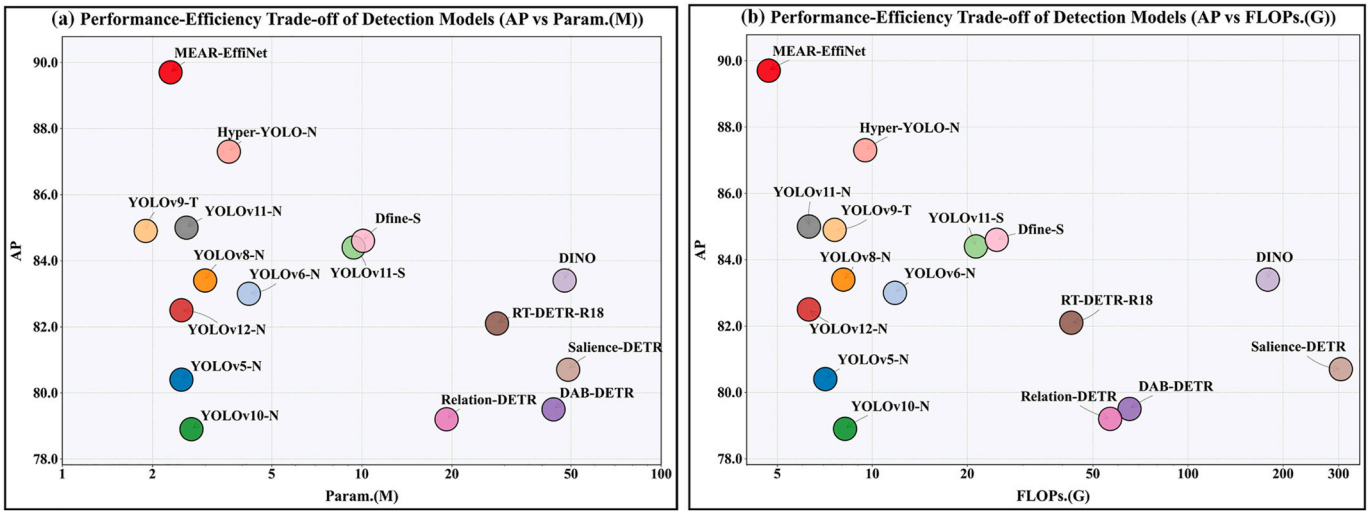

Fig. 11. Performance-efficiency trade-off of detection models (a)AP vs. FLOPs(G) and (b)AP vs. Param. (M).

Table 3
Comparison experiment of different backbone network architectures on MEAR-EffiNet performance. Bold and underline fonts are the highest and the second highest, respectively.

Model	*Param.*(*M*)	*FLOPs*(*G*)	*AP*	AP_{50}	AP_{75}	AP_S	AP_M	AP_L	*Precision*	*Recall*
ConvNeXt V2_pico	5.4	12.6	87.1	55.9	54.6	43.4	56.6	55.8	89.1	77.9
EfficientVit	3.7	7.9	87.8	55.1	55.3	54.3	55.1	54.8	89.4	79.7
FasterNet	3.9	9.2	82.9	51.0	52.1	40.1	42.4	53.0	89.3	74.4
LSKNet	5.6	18.2	88.0	54.7	54.9	53.4	55.9	57.6	77.9	**85.6**
PKINet	5.8	23.6	88.3	55.3	**55.6**	**58.1**	56.7	54.2	86.2	83.2
RepViT	4.5	17.0	85.3	53.9	54.5	38.9	54.6	57.0	88.9	77.4
Ours	**2.3**	**4.7**	**89.7**	**56.1**	55.3	57.6	**57.1**	**59.2**	**94.5**	81.0

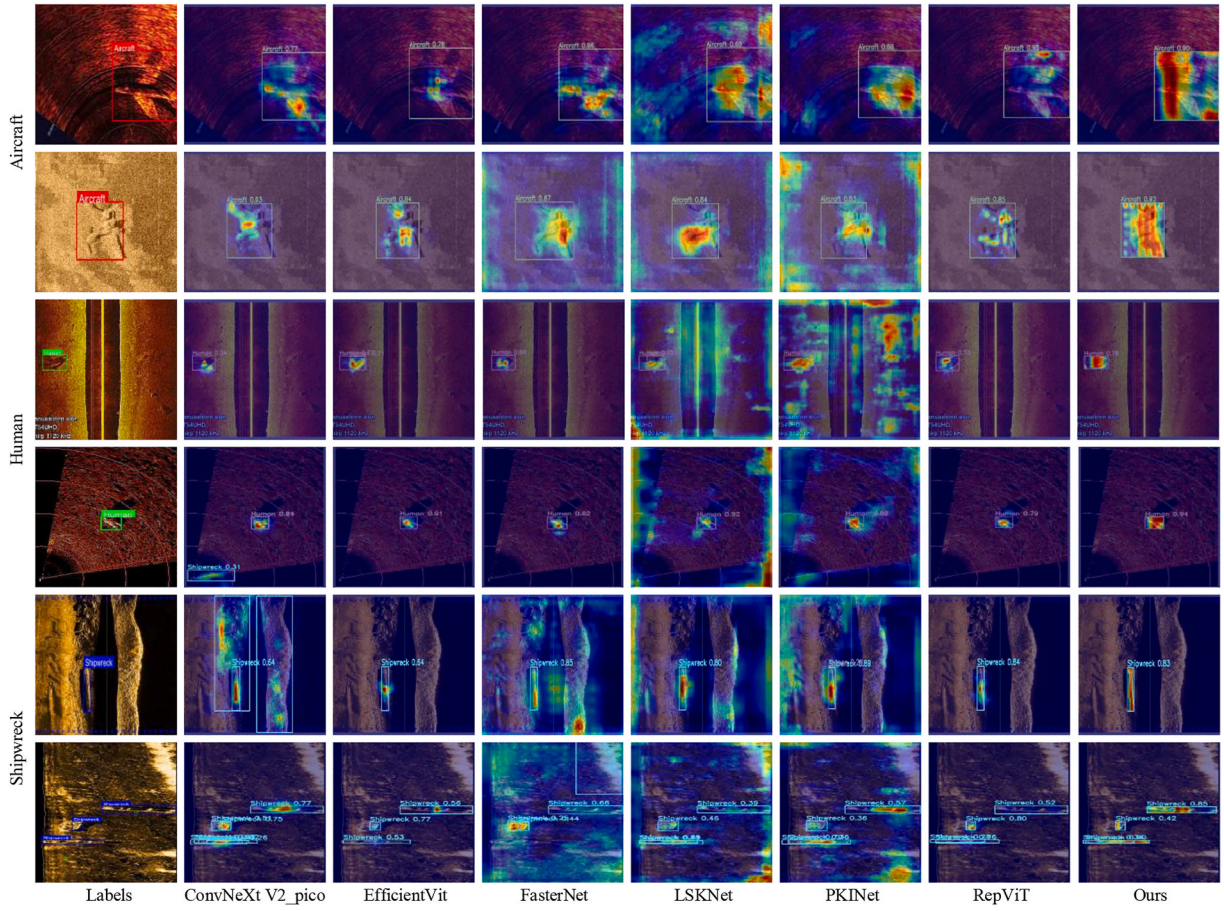

Fig. 12. Comparison of heatmaps generated by different backbone network architectures.

Table 4
Quantitative analysis of feature quality before and after the MSERefiner module at the representative mid-level stage.

Metric	Before	After	Improvement
SNR (dB)	7.84(dB)	8.92 (dB)	+1.07 (dB)
CNR	0.97	1.48	+52.5%

Fig. 13 provides a visual corroboration of these quantitative gains by comparing the feature response heatmaps before and after the MSERefiner module across three typical target categories: Aircraft, Human, and Shipwreck. The first row displays the ground truth labels, while the second and third rows illustrate the feature maps before and after refinement, respectively. In the Before stage (Row 2), the target boundaries appear blurred and exhibit low contrast against the background, with visible noise artifacts interfering with the object structure. In contrast, the After stage (Row 3) reveals a marked transformation: the MSERefiner significantly sharpens the target contours, as evidenced by the high-intensity (red) regions that precisely align with the physical boundaries of the objects. Concurrently, the background regions are effectively suppressed (turning deep blue), indicating that the structural noise has been filtered out. For instance, in the complex Shipwreck and small-scale Human samples, the module recovers distinct shape details that were previously submerged in the background texture, thereby verifying its capability to robustly enhance edges while inhibiting noise.

4.2.4. *Comparison of the MDA-FPN with other feature fusion methods*

To systematically validate the advantages of the proposed MDA-FPN, we conduct comparative experiments by replacing only the feature fusion module while keeping the rest of the network structure, training strategy, and hyperparameters consistent. We compare it with the native FPN from Gold-YOLO (Wang et al., 2023), HSFPN (Chen et al., 2024b), RepGFPN (Xu et al., 2022), and MAFPN (Yang et al., 2024b). The results are shown in Table 5. MDA-FPN achieves an AP of 89.7 with only 2.3 M parameters and 4.7 G FLOPs. Compared to RepGFPN (3.6M/8.2G, 87.5

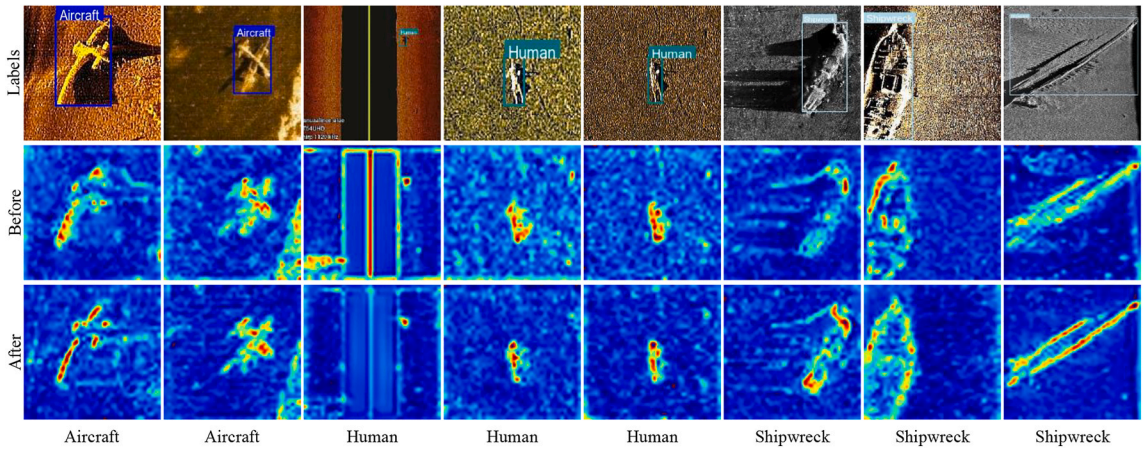

Fig. 13. Visualization of feature response heatmaps before and after the MSERefiner module at the mid-level stage.

Table 5
Comparison experiment of different FPN architectures on MEAR-EffiNet performance. Bold and underline fonts are the highest and the second highest, respectively.

Model	*Param.*(M)	*FLOPs*(G)	*AP*	AP_{50}	AP_{75}	AP_S	AP_M	AP_L	*Precision*	*Recall*
Gold-YOLO	5.9	9.2	84.8	53.4	52.9	37.4	<u>53.6</u>	56.8	90.9	74.7
HSFPN	**1.8**	<u>5.6</u>	85.2	52.4	51.6	54.3	45.1	52.8	83.6	77.4
RepGFPN	3.6	8.2	<u>87.5</u>	**56.2**	53.5	<u>57.1</u>	50.3	56.5	<u>92.0</u>	79.0
MAFPN	2.7	7.1	85.7	55.6	<u>54.1</u>	52.4	49.8	<u>56.6</u>	83.2	**83.6**
Ours	<u>2.3</u>	**4.7**	**89.7**	<u>56.1</u>	**55.3**	**57.6**	**57.1**	**59.2**	**94.5**	<u>81.0</u>

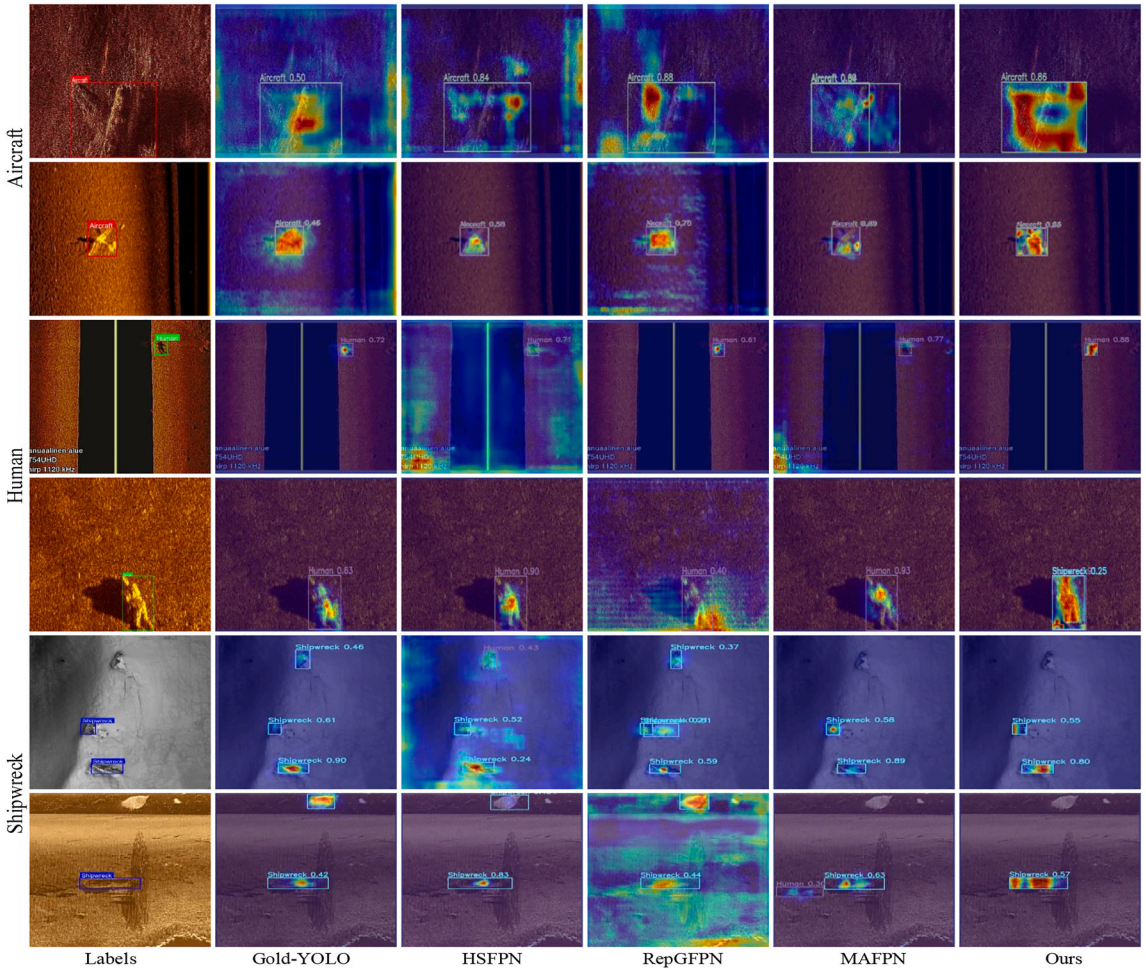

Fig. 14. Comparison of heatmaps generated by different feature fusion network architectures.

AP), its parameter count and computational load are reduced by 36.1% and 42.7%, respectively, while detection accuracy improves by 2.2 percentage points. Compared to HSFPN (1.8 M/5.6G, 85.2 AP), although its parameter count slightly increases, the computational overhead decreases by 16.1%, and the AP improves by 4.5 percentage points.

Visual assessments via heatmaps in Fig. 14 further confirm that MDA-FPN, through its hierarchical focusing and spatially adaptive multi-scale fusion mechanism, enhances semantic interaction and feature expression more effectively than competing methods, achieving an optimal balance between model complexity and performance suitable for resource-constrained scenarios. Moreover, Table 5 shows comprehensive performance metrics, including Precision, Recall, and specific AP scores for small (AP_S), medium (AP_M), and large (AP_L) objects.

4.3. Generalization experiments

4.3.1. Comparison with other methods on the SeabedObjects-KLSG dataset

To assess the generalization robustness of the proposed MEAR-EffiNet across diverse acoustic environments, benchmarking is conducted on the SeabedObjects-KLSG dataset.

As shown in Table 6, on the SeabedObjects-KLSG dataset, MEAR-EffiNet (2.3M Params/4.7G FLOPs) achieves an AP of 68.1%, outperforming the baseline model YOLOv11-N's 59.5% AP while reducing the parameter count and computational cost by 11.5% and 25.4%, respectively. Compared to YOLO series methods known for balancing recognition accuracy and model efficiency, MEAR-EffiNet achieves a 7.7% increase in AP over the current state-of-the-art YOLOv12-N (60.4% AP). Moreover, while being significantly more computationally efficient than Salience-DETR (48.9M Params/305G FLOPs), it still improves the AP by 5.5%. Fig. 15 illustrates the visualization results of different methods on the SeabedObjects-KLSG dataset, highlighting that some targets with blurred edges or extremely small sizes may still suffer from missed detections or false positives. Finally, Table 6 provides a comprehensive breakdown of performance metrics, including Precision, Recall, FPS, and specific AP scores for small (AP_S), medium (AP_M), and large (AP_L) objects.

4.3.2. False positive suppression analysis on negative samples

To evaluate the capacity of the MEAR-EffiNet for false positive suppression in target-free environments, a negative sample test is conducted using 100 negative images from the SeabedObjects-KLSG dataset, covering flat, rippled, and rocky terrains. The false positive rate (FPR) is assessed under two confidence thresholds: a standard setting (Conf = 0.4) and a high-sensitivity setting (Conf = 0.2). As shown in Table 7, quantitatively, at Conf = 0.4, an FPR of 2% is obtained. When the threshold is lowered to Conf = 0.2, an FPR of 11% is observed. These metrics indicate that a low false alarm rate is maintained in most background scenarios while sensitivity is retained.

Fig. 16 presents qualitative results on samples selected to illustrate representative performance behaviors. The first row displays ground truth, while the second and third rows show detections at Conf = 0.4 and Conf = 0.2, respectively. In columns (a)-(c), no false positives (FPs) are generated at either threshold. Columns (d) and (e) show samples where the model remains stable at Conf = 0.4 but triggers false alarms when the threshold is reduced to 0.2, indicating borderline cases. Columns (f) and (g) represent hard negative samples where false positives occur at both thresholds. These errors are primarily attributed to prominent rocks casting acoustic shadows that geometrically resemble valid targets.

4.3.3. Comparison with other methods on the Optical-Ship dataset

We further extend the experiments to the Optical-Ship dataset to evaluate the model's robustness under significant domain shifts. It is important to explicitly state the physical limitations of this dataset: unlike real sonar data, this dataset is constructed via deep transfer learning. It is important to note that these synthetic images fail to reproduce fundamental acoustic physical properties (such as frequency-dependent backscatter intensity and geometric acoustic shadows caused by grazing angles). Consequently, this experiment serves as a generic cross-domain detection test rather than a validation of acoustic-physics modeling.

Despite these limitations, as shown in Table 8, MEAR-EffiNet demonstrates exceptional performance with a 96.9% mAP (IoU = 0.5). Specifically, MEAR-EffiNet achieves a 4.2% increase in AP compared to the current state-of-the-art D-FINE (72.2% AP). Moreover, compared to Relation-DETR (94.4% AP), the proposed model not only improves AP by 2.5% but also significantly reduces the parameter count and computational cost by 88.0% and 91.7%, respectively. These results demonstrate that while the dataset lacks rigorous acoustic fidelity, MEAR-EffiNet possesses strong discriminative power that generalizes well to unseen data distributions and visual styles, maintaining an excellent balance of high accuracy and efficiency. Fig. 17 presents the visualization results of various methods on selected samples from the Optical-Ship dataset. Finally, Table 8 shows comprehensive performance metrics, including Precision, Recall, FPS, and specific AP scores for small (AP_S), medium (AP_M), and large (AP_L) objects.

Table 6
Comparison with state-of-the-art models on SeabedObjects-KLSG. Bold and underline fonts are the highest and the second highest, respectively.

Model	*Param.(M)*	*FLOPs(G)*	*AP*	AP_{50}	AP_{75}	AP_S	AP_M	AP_L	*FPS*	*Precision*	*Recall*
Real − time Object Detectors											
YOLOv5-N	2.5	7.1	61.4	34.5	33.4	17.4	29.9	34.8	68	67.0	61.8
YOLOv6-N	4.2	11.8	62.7	36.6	35.5	18.3	33.6	35.7	**72**	67.4	62.0
YOLOv8-N	3.0	8.1	59.3	33.0	35.4	16.3	29.8	32.5	62	70.8	60.0
YOLOv9-T	**1.9**	7.6	59.3	31.5	27.3	18.7	25.0	32.4	64	63.8	61.4
YOLOv10-N	2.7	8.2	52.4	30.0	28.8	13.2	25.0	31.6	59	62.3	54.8
YOLO11-S	9.4	21.3	62.1	36.5	34.3	19.7	30.8	34.4	34	74.9	63.8
YOLOv12-N	2.5	6.3	60.4	36.0	35.7	20.5	26.7	35.9	62	58.9	61.8
Hyper-YOLO-N	3.6	9.5	67.4	36.4	33.7	**23.5**	31.8	35.3	53	**82.2**	61.7
End − to − end Object Detectors											
DAB-DETR	43.7	65.3	61.2	35.6	36.8	18.5	28.1	35.2	-	63.7	75.3
DINO	47.6	179.0	64.7	34,1	34.6	21.3	24.8	36.7	-	71.3	56.8
RT-DETR-R18	28.3	42.7	61.5	34.9	35.2	19.6	25.4	37.2	-	62.4	66.3
Salience-DETR	48.9	305.0	62.6	37.2	36.2	20.3	26.7	37.4	-	56.1	84.9
Relation-DETR	19.2	56.7	59.5	36.7	37.5	17.9	27.3	37.1	-	54.3	**88.8**
D-FINE-S	10.1	24.8	64.1	37.6	38.4	20.7	30.6	35.3	-	58.7	71.3
Baseline											
YOLO11-N	2.6	6.3	59.5	35.6	38.5	17.4	30.0	33.6	61	61.2	59.9
Ours	2.3	**4.7**	**68.1**	**43.7**	**42.1**	21.3	**36.6**	**40.6**	65	72.1	63.9

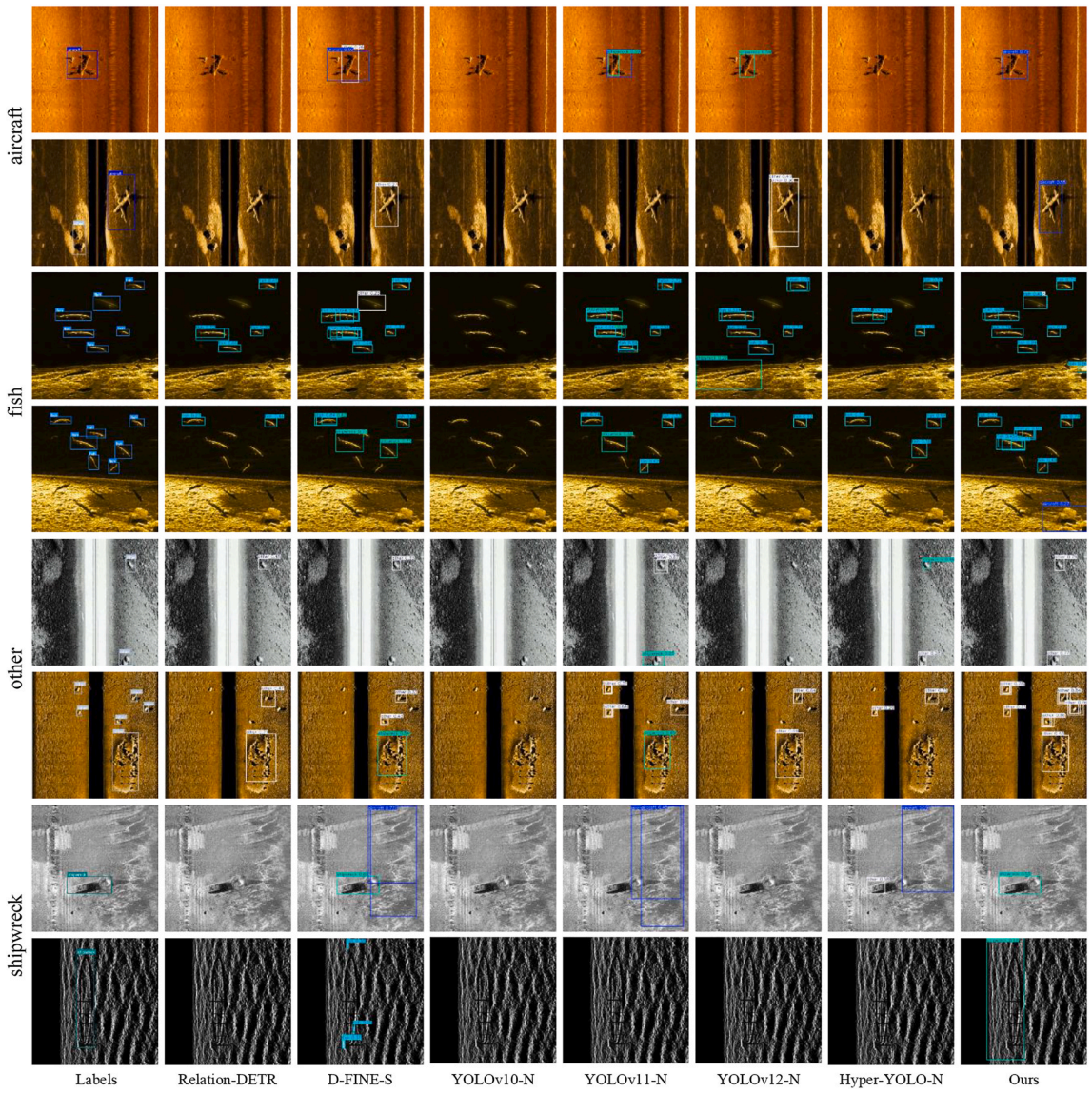

Fig. 15. Visualization results of representative object detection methods for four object categories on the SeabedObjects-KLSG dataset.

Table 7
False positive rate (FPR) analysis.

Metric	Conf	Samples	FPs	FPR (%)
Standard	0.4	100	2	2%
High-Sensitivity	0.2	100	11	11%

4.4. Ablation experiments

4.4.1. Ablation experiments between different components of the overall network structure

To validate the individual and synergistic contributions of each proposed module, comprehensive ablation studies are conducted on the SeabedObject-SSS dataset. As show in Table 9, the baseline model is initialized with 2.6M parameters, 6.3G FLOPs, and a detection accuracy (AP) of 85.0%. First, with the integration of MSERefiner, parameters are reduced by 3.9% (to 2.5M) while AP is boosted to 86.6%. It is confirmed that blurred contours are effectively sharpened and noise is suppressed by its specific edge enhancement capabilities, substantially improving feature representation. Subsequently, AP is elevated to 87.4% with negligible computational overhead by the addition of C2MMRA. It is validated that long-range contextual dependencies are successfully captured by the macro-micro routing mechanism, while subtle local structures are extracted from complex acoustic backgrounds.

Moreover, the incorporation of MDA-FPN optimizes the computational load to 5.8G (a 7.9% reduction) and significantly raises AP to 89.1%, highlighting its ability to aggregate multi-scale features efficiently for holistic image processing. Finally, deploying the HARDEN head reduces parameters and FLOPs by 7.7% and 17.5% (to 2.3M and 4.7G), respectively, while maximizing AP to 89.7%. This demonstrates HARDEN's superior robustness against noise and precision in localization. Fig. 18 visually summarizes these dual advantages in accuracy and lightweight design.

Additionally, Table 9 provides a comprehensive breakdown of performance metrics, including Precision, Recall, and specific AP scores for small (AP_S), medium (AP_M), and large (AP_L) objects.

4.4.2. Ablation experiments between different components in MDA-FPN

To systematically validate the efficacy of FocSALA and C3CFA within the MDA-FPN architecture, ablation studies are conducted, with quantitative results show in Table 10. An AP of 87.9% with 2.3M parameters and 5.1G FLOPs is yielded by the baseline configuration (excluding both modules). The computational load is significantly reduced by 11.8% (to

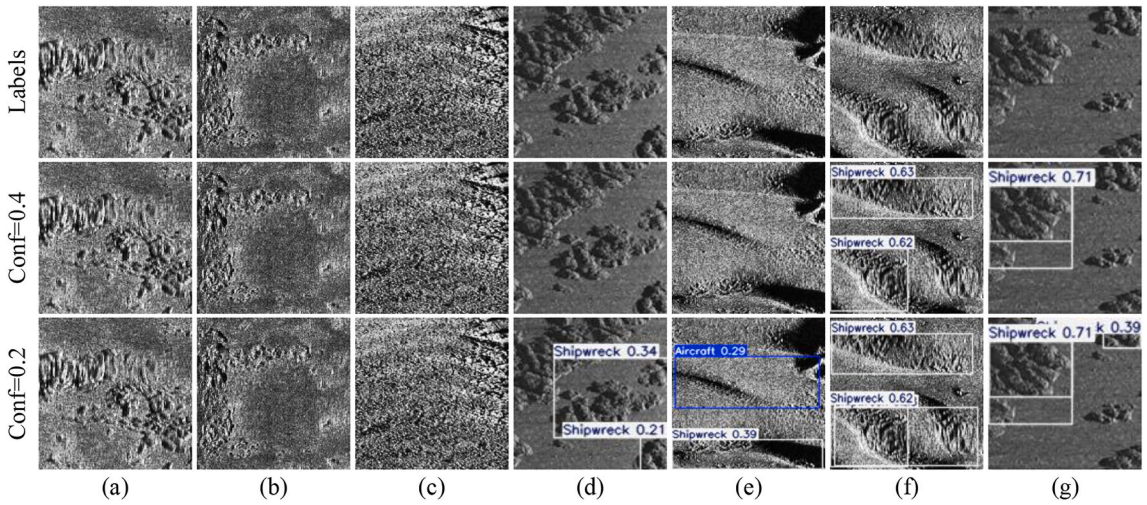

Fig. 16. Visualization of detection results on negative samples under different confidence thresholds.

Table 8
Comparison with state-of-the-art models on Optical-Ship. Bold and underline fonts are the highest and the second highest, respectively.

Model	*Param.*(*M*)	*FLOPs*(*G*)	*AP*	AP_{50}	AP_{75}	AP_S	AP_M	AP_L	*FPS*	*Precision*	*Recall*
Real − time Object Detectors											
YOLOv5-N	2.5	7.1	93.1	61.7	71.4	70.0	59.3	61.9	75	93.9	92.5
YOLOv6-N	4.2	11.8	91.8	65.6	71.9	70.0	60.2	66.0	**79**	93.3	92.6
YOLOv8-N	3.0	8.1	93.9	65.3	74.0	60.0	61.5	65.6	68	95.9	92.2
YOLOv9-T	**1.9**	7.6	94.6	65.9	72.6	60.0	62.3	66.4	73	93.8	93.7
YOLOv10-N	2.7	8.2	93.7	63.4	69.6	70.0	52.6	64.2	65	94.4	90.0
YOLO11-S	9.4	21.3	93.7	65.5	73.9	60.0	62.4	65.7	40	95.2	92.4
YOLOv12-N	2.5	6.3	93.8	59.3	63.2	70.0	53.6	59.9	69	95.4	90.7
Hyper-YOLO-N	3.6	9.5	95.1	64.5	73.0	80.0	50.7	65.5	61	95.0	92.7
End − to − end Object Detectors											
DAB-DETR	43.7	65.3	91.2	61.5	65.7	42.8	61.3	63.5	-	92.8	91.5
DINO	47.6	179.0	94.2	64.6	70.2	47.1	60.7	64.5	-	93.8	93.4
RT-DETR-R18	28.3	42.7	93.1	65.3	67.2	37.8	56.4	64.3	-	93.5	92.8
Salience-DETR	48.9	305.0	93.4	62.2	69.2	52.9	59.2	62.8	-	90.5	**97.8**
Relation- DETR	19.2	56.7	94.4	63.3	68.9	28.2	54.7	64.2	-	91.4	**97.8**
D-FINE-S	10.1	24.8	94.6	65.1	72.3	46.8	61.3	65.1	-	94.9	94.1
Baseline											
YOLO11-N	2.6	6.3	94.3	65.3	73.3	80.0	**65.9**	65.2	68	95.4	93.4
Ours	2.3	**4.7**	**96.9**	**70.4**	**76.5**	**90.0**	61.2	**67.2**	75	**96.3**	92.4

4.5G) by introducing FocSALA alone, while AP is elevated to 88.6% (+0.8%). FocSALA's role as a lightweight, spatially-aware enhancement mechanism is underscored, wherein adaptive pooling is utilized to robustly suppress noise and handle multi-scale data with minimal overhead.

Conversely, deploying only the C3CFA module results in an AP of 88.2%, with a slight computational increase to 5.3G. Despite the marginal cost, its global-local dual-path processing proves vital for extracting discriminative features from complex SSS signals, particularly for identifying targets under low-SNR or partial burial conditions. Crucially, the full integration of both modules achieves a peak AP of 89.7% (+2.1%) while optimizing FLOPs to 4.7G (a 7.8% reduction), significantly outperforming individual configurations. This demonstrates a potent synergistic effect: FocSALA provides efficient spatial localization and noise suppression, while C3CFA delivers fine-grained multi-dimensional feature refinement. Their combination effectively reduces redundant computations and enhances discriminability against acoustic artifacts, ensuring robust adaptability in diverse underwater environments.

Finally, Table 10 shows comprehensive performance metrics, including Precision, Recall, and specific AP scores for small (AP_S), medium (AP_M), and large (AP_L) objects.

4.5. Discussion and future work

Based on the comprehensive experimental analysis, superior detection accuracy and computational efficiency are demonstrated by MEAR-EffiNet. However, the difference between the object-centric SeabedObjects-KLSG dataset and continuous survey-line imagery encountered in real-world missions must be noted. The datasets utilized in this study are primarily object-centric, where images are pre-cropped to feature targets prominently. In contrast, real-world marine surveys involve continuous survey-line imagery, characterized by vast areas of featureless seabed background, extreme aspect ratios, and sparse target distribution. Therefore, this domain gap is explicitly acknowledged, and the current validation is clarified as a preliminary step prior to application in complex, large-scale survey environments.

To bridge this gap and strengthen practical engineering relevance, future work will focus on validating the model against authentic measurement datasets. Specific attention will be given to engineering geological datasets such as the Bohai Sea engineering geological dataset (Marine-PULSE) (Du et al., 2023; Du et al., 2025). These datasets provide the continuous, raw acoustic logs necessary to evaluate the model's false alarm rate and scanning efficiency in actual missions. Furthermore, physics-informed priors will be integrated for extreme environments, and the model will be further optimized for deployment on

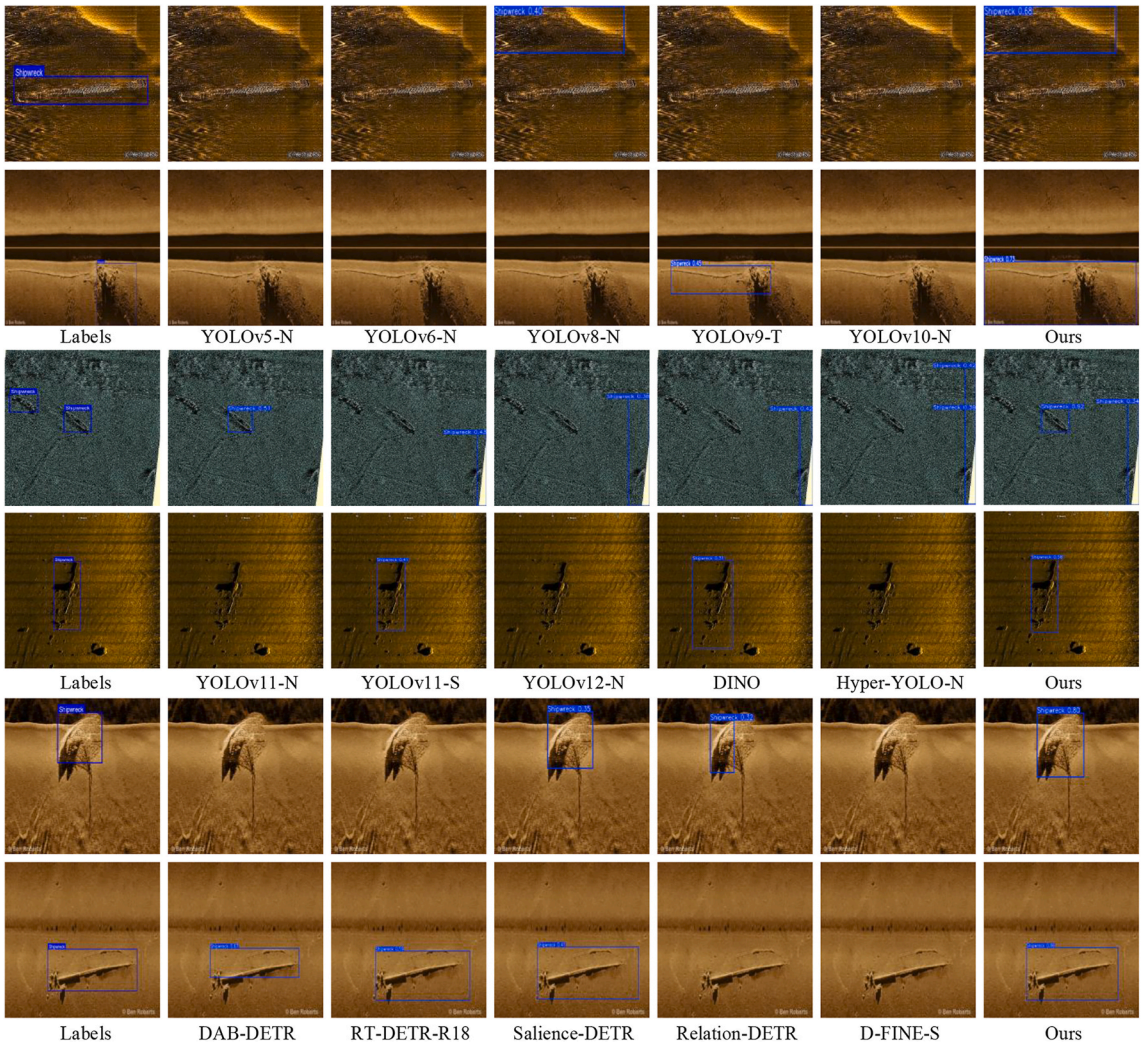

Fig. 17. Qualitative results of representative object detection methods on the Optical-Ship dataset.

Table 9
Ablation experiments between different components of the overall network structure on SeabedObject-SSS. Bold and underline fonts are the highest and the second highest, respectively.

MSER-Net	C2MMRA	MDA-FPN	HARDEN	***Param.*** (***M***)	***FLOPs*** (***G***)	***AP***	AP_{50}	AP_{75}	AP_S	AP_M	AP_L	***Precision***	***Recall***
				2.58	6.3	85.0	53.7	52.4	25.0	**58.2**	54.6	91.2	77.5
√				2.53	6.3	86.6	54.3	55.6	37.6	57.4	55.4	93.2	79.6
√	√			2.54	6.3	87.4	55.8	**60.2**	25.7	56.8	**62.3**	92.7	82.6
√	√	√		2.50	5.8	89.1	55.6	55.9	50.0	56.2	54.7	89.8	**83.4**
√	√	√	√	**2.27**	**4.7**	**89.7**	**56.1**	55.3	**57.6**	57.1	59.2	**94.4**	81.0

computation-constrained edge devices.

5. Conclusion

MEAR-EffiNet is experimentally presented for underwater SSS object detection, employing a synergistic multi-module architecture. The MEAR-EffiNet includes four core modules, MSERefiner, C2MMRA, MDA-FPN, and HARDEN, which are detailed to address specific SSS image challenges. These four modules collaborate to progressively enhance and detect targets. Each module is tailored for tasks ranging from edge refinement and context capture to robust localization. Specifically, blurred edges and multi-scale details are enhanced by this multi-module architecture. Semantic discriminability in complex backgrounds with low SNR is improved. Detection accuracy is balanced with computational efficiency. Moreover, we design the MDA-FPN neck and the HARDEN detection head, which synergistically leverage acoustic textures and model edge uncertainty to ensure precise target localization. Extensive experiments on the SeabedObject-SSS dataset and various complex SSS scenarios (e.g., sunken aircraft, human bodies, shipwrecks) verify the proposed method. Competitive accuracy and efficiency are achieved by MEAR-EffiNet, outperforming recent state-of-the-art models like the YOLO and DETR series. This work establishes a solid foundation for real-time, high-precision underwater perception tasks.

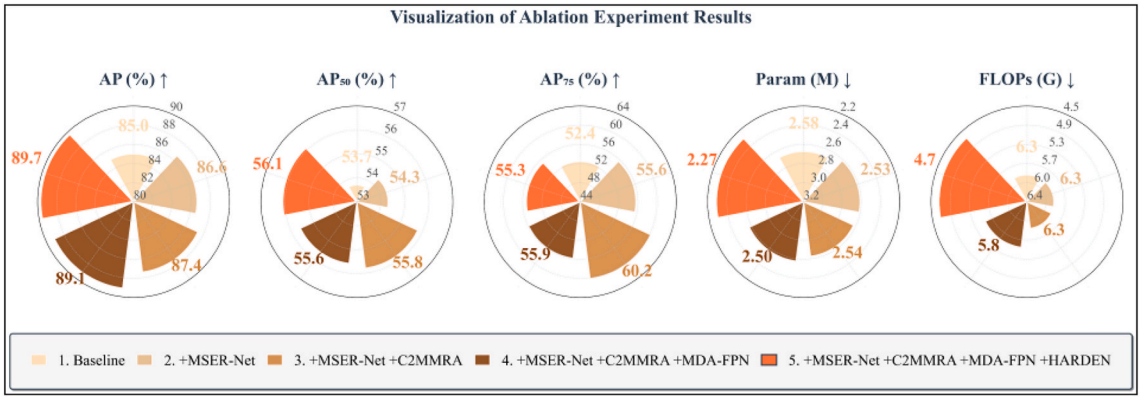

Fig. 18. Visualization of ablation experiment results.

Table 10
Ablation experiments between different components in MDA-FPN on SeabedObject-SSS. Bold and underline fonts are the highest and the second highest, respectively.

FocSALA	C3CFA	*Param.*(*M*)	*FLOPs*(*G*)	*AP*	AP_{50}	AP_{75}	AP_S	AP_M	AP_L	*Precision*	*Recall*
		2.3	5.1	87.9	54.7	58.5	47.6	57.1	55.6	85.8	80.0
√		2.3	**4.5**	88.6	**56.5**	**60.9**	36.8	56.5	**59.4**	90.2	80.9
	√	2.3	5.3	88.2	55.4	54.8	35.2	55.4	57.4	92.8	**84.0**
√	√	**2.3**	4.7	**89.7**	56.1	55.3	**57.6**	**57.1**	59.2	**94.4**	81.0

CRediT authorship contribution statement

Zijian Zeng: Writing – review & editing, Writing – original draft, Software, Resources, Project administration, Methodology, Data curation. **Genping Zhao:** Validation, Investigation, Data curation, Conceptualization. **Heng Wu:** Visualization, Validation, Supervision, Investigation, Funding acquisition, Data curation, Conceptualization. **Chunhua He:** Supervision, Investigation, Data curation, Conceptualization. **Huapan Xiao:** Investigation, Funding acquisition, Formal analysis, Data curation.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

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